

A project report on

IOT BASED HOME AUTOMATION SYSTEM

Submitted in partial fulfilment for the award of the degree of

Bachelor of Computer Science

By

S. VIVEKANANATHAN (21BCS0078)

K. DHANUSH (21BCS0166)



VIT[®]
Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

**SCHOOL OF COMPUTER SCIENCE ENGINEERING AND
INFORMATION SYSTEMS**

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DECLARATION

I hereby declare that the thesis entitled "IOT BASED HOME AUTOMATION SYSTEM" submitted by me, for the award of the degree of B.Sc. (Computer Science) is a record of bonafide work carried out by me under the supervision of Dr. SENTHIL KUMAR N

I further declare that the work reported in this thesis has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.

Place: Vellore

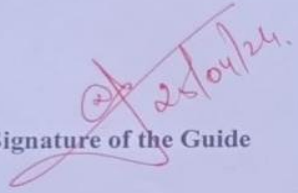
Date: 25-04-2024

S. Vinodh
b. Senthil
Signature of the Candidate

CERTIFICATE

This is to certify that the thesis entitled "IOT BASED HOME AUTOMATION SYSTEM" submitted by S. VIVEKANANTHAN (21BCS0078), K.DHANUSH (21BCS0166), School of Computer Science Engineering and Information Systems, Vellore Institute of Technology, Vellore for the award of the degree B.Sc. (Computer Science) is a record of bonafide work carried out by him/her under my supervision.

The contents of this report have not been submitted and will not be submitted either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university. The Project report fulfils the requirements and regulations of VELLORE INSTITUTE OF TECHNOLOGY, VELLORE and in my opinion meets the necessary standards for submission.


Signature of the Guide

Signature of the HOD

Internal Examiner

External Examiner

ABSTRACT

This project presents an IoT-based home automation system designed to enhance convenience, security, and energy efficiency in residential environments. Leveraging the capabilities of the ESP32 microcontroller, the system allows users to remotely control lighting, fan, and door access through a user-friendly mobile application developed using Flutter. Access to the door is secured by a username and password authentication mechanism, ensuring privacy and safety. Real-time monitoring of temperature and electrical usage is facilitated through the mobile application, providing users with insights into energy consumption patterns. Additionally, the system integrates webcam functionality using Yawcam software, enabling users to remotely monitor their home environment via a laptop. Firebase authentication and real-time database services are utilized for seamless data transmission between the mobile application and the microcontroller. To enhance security further, YOLOv8 object detection is implemented to detect human presence in the room, triggering automatic activation of lights and fans. Data exchange between devices is facilitated by the MQTT protocol through the Mosquitto broker. Furthermore, an Arduino Uno board is incorporated to detect gas leakage using an MQ5 sensor, activating a buzzer alarm for prompt response. Additionally, a night light functionality controlled by a photoresistor enhances safety during nighttime. Overall, this comprehensive home automation system offers a seamless and secure solution for remote monitoring and control, contributing to enhanced comfort, security, and energy efficiency in modern households.

ACKNOWLEDGEMENT

It is my pleasure to express with deep sense of gratitude to Prof. Senthil Kumar N, Assistant Professor Sr. Grade 2, School of Information Technology and Engineering and Information System, Vellore Institute of Technology, for his constant guidance, continual encouragement, understanding; more than all, he taught me patience in my endeavour. My association with him is not confined to academics only, but it is a great opportunity on my part of work with an intellectual and expert in the field of computer technology.

I would like to express my gratitude to DR. G. VISWANATHAN, Chancellor, VIT, VELLORE, MR. SANKAR VISWANATHAN, DR. SEKAR VISWANATHAN, MR .G V SELVAM, Vice – Presidents VIT, VELLORE, Dr. V. S. Kanchana Bhaaskaran, I/c Vice – Chancellor, Dr. Partha Sharathi Mallick, Pro-Vice Chancellor and Dr. S. Sumathy, Dean, School of Computer Science Engineering And Information Systems, for providing with an environment to work in and for his inspiration during the tenure of the course.

In jubilant mood I express ingeniously my whole-hearted thanks to Dr. Iyyaparaja M, HOD/Professor, all teaching staff and members working as limbs of our university for their not-self-centred enthusiasm coupled with timely encouragements showered on me with zeal, which prompted the acquirement of the requisite knowledge to finalize my course study successfully. I would like to thank my parents for their support.

It is indeed a pleasure to thank my friends who persuaded and encouraged me to take up and complete this task. At last, but not least, I express my gratitude and appreciation to all those who have helped me directly or indirectly toward the successful completion of this project.

Place: Vellore

Name of the students:

Date:

VIVEKANANTHAN.S – 21BCS0078

DHANUSH.K– 21BCS0166

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- YOLO - You Only look Once.
- MQTT – Message Queuing Telemetry Transport
- HTC – Hardware Test case
- STC – Software Test Case
- TC – Test case

CHAPTER 1

INTRODUCTION

1.1 BACKGROUND

Recent advancements in technology, especially in IoT, have transformed home automation systems. These systems streamline tasks, enhance energy efficiency, and boost security. Utilizing microcontrollers, wireless communication, and mobile apps, modern home automation allows remote control over various devices. Integration of machine learning like YOLOv8 enables intelligent decision-making and security measures. Cloud-based services such as Firebase ensure secure data storage and real-time communication. Sensors like gas detectors and photoresistors add safety and functionality. Overall, these advancements make home automation more accessible, user-friendly, and efficient.

1.2 MOTIVATION

The motivation behind this project is twofold education and hands-on learning about IoT technology. By undertaking this project, participants can immerse themselves in the intricacies of IoT systems, gaining practical experience with microcontrollers, sensors, wireless communication protocols, and cloud services. This project serves as a valuable educational tool, offering students and enthusiasts an opportunity to understand the fundamentals of IoT and explore its applications in real-world scenarios. Moreover, by developing a comprehensive home automation system, participants can deepen their understanding of smart home technology and its potential impact on modern living. Through experimentation and problem-solving, individuals can hone their technical skills, foster creativity, and cultivate a curiosity-driven approach to learning, laying the foundation for future innovation in the field of IoT.

1.3 PROBLEM STATEMENT

- **Demand for Efficient Home Automation:** There is a growing demand for home automation systems that offer convenience, energy efficiency, and security.
- **Lack of Seamless Integration:** Traditional home automation solutions often lack seamless integration, leading to disjointed user experiences and limited functionality.

- **Need for Intuitive Interfaces:** Users require intuitive interfaces that allow for easy control and monitoring of home devices, without the need for complex setups or technical expertise.
- **Importance of Robust Security Measures:** Ensuring the security and privacy of home automation systems is crucial to protect against unauthorized access or malicious attacks.
- **Educational Opportunity:** There is a need for educational resources that provide hands-on experience in designing and implementing IoT-based systems, offering students and enthusiasts a practical platform for learning.
- **Integration of Various Components:** The challenge lies in integrating various components, including microcontrollers, sensors, mobile applications, and cloud services, to create a reliable and user-friendly home automation solution.
- **Empowering Learning and Innovation:** By providing a comprehensive home automation system and educational resources, the project aims to empower individuals to explore the potential applications of IoT technology and contribute to future innovations in the field.

1.4 OBJECTIVES

- **Develop a Comprehensive Home Automation System:** Create a fully functional home automation system that integrates seamlessly with various household devices such as lights, fans, doors, and sensors.
- **Implement Remote Control Functionality:** Enable users to remotely control home devices through a mobile application developed using Flutter, allowing for convenient access from anywhere with internet connectivity.
- **Ensure Secure Access Control:** Implement a username and password authentication mechanism to ensure secure access to the door-opening feature, enhancing privacy and safety.
- **Enable Real-time Monitoring:** Integrate sensors to monitor temperature, current usage, and gas detection, providing users with real-time insights into their home environment via the mobile application.
- **Facilitate Remote Surveillance:** Incorporate webcam functionality using Yawcam software, allowing users to monitor their home environment remotely from a laptop.
- **Utilize Firebase for Data Transmission:** Utilize Firebase authentication and real-time database services for secure and seamless data transmission between the mobile application and the microcontroller.

- **Implement Object Detection for Automation:** Integrate YOLOv8 object detection to detect human presence in the room, triggering automatic activation of lights and fans for enhanced convenience and energy efficiency.
- **Utilize MQTT Protocol for Data Exchange:** Facilitate data exchange between devices using the MQTT protocol through the Mosquitto broker, ensuring efficient communication and synchronization.
- **Enhance Safety with Gas Detection and Night Light:** Incorporate gas detection using an MQ5 sensor and activate a buzzer alarm in case of gas leakage. Implement a night light functionality using a photoresistor for enhanced safety during nighttime.
- **Provide a Learning Platform:** Offer a practical platform for individuals to learn about IoT technology, mobile application development, cloud services, and security measures through hands-on experience with the project implementation.

1.5 SCOPE OF THE PROJECT

- **Hardware Implementation:** The project will involve the setup and configuration of hardware components, including microcontrollers (ESP32, Arduino Uno), sensors (temperature, gas, photoresistor), and a webcam for remote surveillance.
- **Software Development:** Development of software components will include creating a mobile application using Flutter for remote control and monitoring, implementing object detection using YOLOv8, integrating Firebase for authentication and real-time data exchange, and configuring Yawcam for webcam functionality.
- **Functionality:** The project will focus on implementing core functionalities such as remote control of lights, fans, and door access through the mobile application, real-time monitoring of temperature, electrical usage, and gas detection, automated activation of lights and fans based on human presence detection, and alerting users in case of gas leakage.
- **Security Measures:** Implementation of security measures will include username and password authentication for door access, encryption of data transmission between devices, and ensuring secure access to the system to prevent unauthorized usage.
- **Testing and Validation:** Rigorous testing and validation will be conducted to ensure the reliability, functionality, and security of the home automation system. This will involve both unit testing of individual components and integration testing of the system.

- **Documentation and Knowledge Sharing:** Comprehensive documentation will be provided to guide users through the setup, configuration, and usage of the home automation system. Additionally, knowledge sharing sessions or workshops may be conducted to share insights and lessons learned from the project with the broader community.
- **Educational Component:** The project will serve as a valuable educational resource for individuals interested in learning about IoT technology, mobile application development, cloud services, and security measures. Participants will have the opportunity to gain hands-on experience and deepen their understanding of these concepts through practical implementation.
- **Scalability and Future Expansion:** While the initial scope of the project will focus on implementing core functionalities, the system will be designed to be scalable and easily expandable to accommodate additional features or devices in the future, allowing for continuous improvement and customization based on user feedback and evolving technology trends.

CHAPTER 2

LITERATURE SURVEY

2.1 SUMMARY OF THE EXISTING WORK

S.NO	AUTHORS	PAPER TITL, DOP, JOURNAL	Techniques Followed	Advantages
[1]	Joseph, Stephen Bassi, Joseph, S. B., Dada, E. G., Yakubu, J. H., & Moses, E. I.	Design and Modelling of Smart Home with Internet of Things Enabled Automation and Control System. COVENANT JOURNAL OF ENGINEERING TECHNOLOGY (2023).	Development of IoT-based smart home automation. Real-time monitoring and equipment management. Scalability with additional sensors and actuators. Utilization of solar panels for power. Printed circuit boards for optimized circuitry.	Portability and user-friendliness. Cost-effectiveness. Enhanced environmental friendliness. Improved energy efficiency. Reliability and reduced risk of malfunctions.
[2]	Anuradha, P., et al, Anuradha, P., Vasanth, K., Renuka, G., & Rao, A. R, Anuradha.	IoT based enabling home automation system for individuals with diverse disabilities. e-Prime-Advances in Electrical Engineering, Electronics and Energy 6 (2023): 100366.	Integration of autonomous industry machine controlling. Utilization of ESP32 for gesture recognition. Implementation of Node MCU for speech recognition. Incorporation of app-based button controls connected to IoT.	Simplification and enhancement of daily life. Catering to users of all ages and abilities, overcoming demographic limitations. Versatile and convenient machine control methods: voice control, app-based button interactions, and

				gesture recognition. Contribution to energy efficiency. Automatic shutdown of machines and appliances left inadvertently on, enhancing security, and promoting a greener lifestyle. Improved convenience, efficiency, and security in daily life. Potential to redefine interactions with surroundings and elevate overall quality of life.
[3]	Arman, Syed Mohammad Sohel	Iot Based Home Automation and Security. Department of Electronic and Telecommunication Engineering, 2023.	Technological innovation is crucial for scientific development in the current industrial revolution. Our project effectively applies technological innovation and will benefit automated home automation techniques.	Our solution achieves intended objectives with good accuracy at a low cost compared to competing initiatives. Previous systems were more expensive while offering similar functionalities. Earlier systems were focused on specific themes such as controlling appliances through an app only, whereas our system

				demonstrates a variety of functionalities at a lower cost.
[4]	Ekram, Rashid, Anmol Deshwal, and Anup Yadav	Cloud Based IoT Platforms for Home Automation System. No. 11211. EasyChair, 2023.	Integration of a cloud based IoT platform into a smart home automation system. Utilization of user interface design to enhance user experience. Implementation of remote-control functionalities. Incorporation of AI and ML algorithms to improve system functionality and user experience. Effective implementation of data privacy and security measures.	Significant enhancements in user experience and remote-control functionality. Improved convenience and efficiency within smart homes. Collective improvements in user experience through user interface design, remote control functionalities, and AI/ML algorithms. Enhanced data privacy and security measures ensuring protection of user data. Reduction of manual interaction through the system's ability to predict user preferences and automate actions, thereby enhancing user comfort and convenience.
[5]	Deng, Qizhang	A Modular Home Automation System Based on IoT Technology. Diss. Instytut	Utilization of IoT technologies, specifically cheap	Cost-effectiveness achieved by cheap ESP8266 boards.

		Telekomunikacji, 2023.	ESP8266 boards. Integration of software libraries for Wi-Fi communication and MQTT protocol. Development of a Smart Home model that functions efficiently without requiring a full-time internet connection.	Flexibility in communication through Wi-Fi and MQTT protocol, enabling efficient data transfer. Independence from a full-time internet connection, enhancing reliability and accessibility. Scalability and optimization of the system through recommendations for expansion. Integration of additional devices such as monitors, keypads, and sensors via I2C communication. Long-distance operation facilitated using GSM shield, enabling control via mobile phones or other communication tools.
[6]	Mohamed, Khalil, and Ayman El Shenawy.	A Smart IoT-Based Home Automation System for Controlling and Monitoring Home Appliances. 2023	Integration of IoT, cloud computing, and smart devices to create a comprehensive Home Automation System (HAS).	Well-designed and comprehensive solution catering to various user needs. Wide range of features and functionalities

			Flexibility of the system for implementation in various settings, including residential homes, offices, and businesses. Customizable nature of the system to tailor to the specific requirements of individual users. Incorporation of multiple control methods such as manual operation, remote control, and IoT cloud-based control.	offered by the integration of IoT, cloud computing, and smart devices. Personalized experience ensured by the system's customizable nature. Prominent ease of use enhancing user convenience and accessibility. Versatility in control methods enhancing user convenience and accessibility.
[7]	Wu, Yi Hua, and Hongli Luo.	Apply edge intelligence to IoT based home automation. In 17th Int. Conf. on Applied Computing (2020).	Integration of machine learning (specifically deep learning) at the edge device in an IoT-based home automation system. Utilization of edge computing to enhance the quality of service of the IoT system without relying on an internet connection. Implementation of deep learning techniques for object detection applications. Utilization of	Improvement of service quality in IoT applications through the application of machine learning at the edge. Enhanced performance and responsiveness of the IoT system without dependency on internet connectivity. Capability for touchless interaction with physical devices, enhancing user convenience and accessibility.

			TensorFlow Lite for implementing voice control and gesture recognition on microcontrollers.	
[8]	Pravalika, V., and Ch Rajendra Prasad.	Internet of things-based home monitoring and device control using Esp32. International Journal of Recent Technology and Engineering 8.1S4 (2019): 58-62.	Utilization of IoT to monitor and control multiple devices at home. Improvement of device performance through automatic and remote control. Three-phase system operation: monitoring, automatic control, and user-controlled operation. Integration with cloud platforms (ThingSpeak) and mobile applications (MIT App Inventor) for data storage and remote access.	Enhanced performance of various devices through automatic and remote-control capabilities. Real-time monitoring of temperature, gas leakage, water level, and person detection. Automatic control of motor pump and gas knob based on predefined thresholds. User-friendly mobile app interface for controlling home appliances. Low-cost implementation using embedded microcontroller with Wi-Fi module ESP32. Simplified operation and easy integration with existing home appliances.
[9]	Rathour, Navjot	Sigma Home: An IoT-Based Home Automation Usi	Utilization of IoT for remotely monitoring and	Improved energy efficiency through

	Monika Kumar, Vivek Kundu, Sankha Subhra Gehlot, Yash Gurung, Aditya	ng Node MCU. 2nd International Conference on Edge Computing and Applications (ICECAA) 2023.	managing various home appliances and gadgets. Integration of Node MCU, an open-source Wi-Fi-enabled microcontroller, with IoT for simple and easy network interfacing in home automation. Use of different sensors to monitor home devices such as lights, temperature, humidity, and fan. Data monitoring through an Android application or web-based interface for energy efficiency and electrical fixture control via IoT. Fetching sensor data by the cloud for further analysis and execution of commands to achieve desired results.	remote control of electrical fixtures. Flexibility and resourcefulness in IoT-based home automation. Enhanced quality of life with minimal maintenance, cost, energy consumption, and increased security. Scalability of the system to accommodate future needs. Reduction of human effort through automation of home tasks. Accessibility to laymen, enabling easy adaptation to upcoming technology.
[10]	Mashiat, Lamisa Azad, Sumaiya Binte Jinia, Sadia Naj Islam, Md.	A comprehensive IoT-Based Home Automation Solution: Analyzing Security Threats, Maintenance and Proposal. IEEE Conference on Interdisciplinary Approaches in Technology and Management for Social	Utilization of IoT as a key component of technology in the 4th industrial revolution era. Integration of multiple sensors into a globally connected network enabled by IoT.	Safety enhancement through an authentication mechanism utilizing biometric technology to prohibit unauthorized access. Advanced sensor

	Ariful	Innovation (IATMSI) 2022.	Real-time data updates and preservation of information facilitated by IoT. Design of a biometric technology coupled with a sensor-integrated framework for defense.	module for immediate action upon detection of accidents, ensuring timely response and safety. Web panel server for tracking real-time updates of the residence, providing comprehensive monitoring and control capabilities. Advancement of technology to maintain safety and security of loved ones.
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2.2 CHALLENGES PRESENT IN EXISTING SYSTEM

- **Complex Setup:** Existing home automation systems often require complex setups involving multiple devices and protocols, which can be intimidating for users with limited technical knowledge.
- **Integration Complexity:** Existing home automation systems often present challenges in integrating various components, such as sensors, actuators, and control interfaces, leading to a complex setup process and interoperability issues.
- **User Interface Limitations:** Many systems lack intuitive user interfaces, hindering effective control and monitoring, particularly through mobile applications, as outlined in your project's emphasis on user-friendly mobile app control.
- **Security Concerns:** Security vulnerabilities pose a significant risk in home automation systems, particularly concerning remote access features like door control via a mobile app, highlighting the importance of robust authentication mechanisms and secure data transmission.

- **Scalability and Expansion:** Some systems struggle with scalability, making it challenging to expand the automation setup or integrate new features seamlessly, as your project aims to address through considerations for scalability and future expansion.
- **Reliability and Stability:** Ensuring reliability and stability is crucial, especially in systems that involve critical functionalities like gas detection and automated control of lighting and fans, emphasizing the need for robust hardware and software solutions.
- **Cost and Accessibility:** The high cost associated with purchasing and installing home automation devices can act as a barrier to adoption, underscoring the importance of cost-effective solutions that leverage readily available hardware and open-source software frameworks like ESP32 and Arduino Uno.

CHAPTER 3

REQUIREMENTS

3.1 HARDWARE REQUIREMENTS

- ESP 32 (Microcontroller),
- Arduino UNO (Microcontroller),
- MQ5 Sensor (Gas Sensor),
- PIR Motion Sensor,
- DS18B20 (Temperature Sensor),
- ACS712 20A (Current Sensor),
- Servo Motor
- Relay 5v,
- Buzzer,
- Photoresistor
- Led,
- Bulb,
- 5v Fan,
- Battery 3.3 v lithium ion with holder,
- Jumper wire,
- Wire,
- Junction Box,
- Breadboard,
- Type B cable,
- Arduino cable,
- Laptop.

3.2 SOFTWARE REQUIREMENTS

- Arduino IDE,
- VS Code,
- Flutter SDK,
- Firebase,
- Python,
- OpenCV,
- Paho-mqtt,
- YOLO v8 (model- weight: yolov8n.pt),
- Yawcam (To live stream),
- MQTT Mosquitto,
- MQTT Explorer,
- Cirkkit Designer.

3.3 BUDGET

Procured Items/Components for the Project work	Total Cost
Arduino UNO rev3 (with cable)	720
ESP 32	430
MQ-5	120
PIR Motion Sensor	150
DS18B20 (temperature sensor)	60
ACS712 20A (Current sensor)	175
Servo Motor	110
Relay 5v (75) (x3)	225
Buzzer	25
Breadboard	65
5v Fan	150
led	10
Wire	50
Jumper wire (one 2rs)	(approx.)240
Solder wire	20
Type B cable	149
Bulb (x2)	120
Photoresistor	6
Multimeter	390
Lithium-ion battery and holder (100rs one)	200
Total cost of used product (Rs)	3315

Additional Cost with failure device cost:

For Experiment:

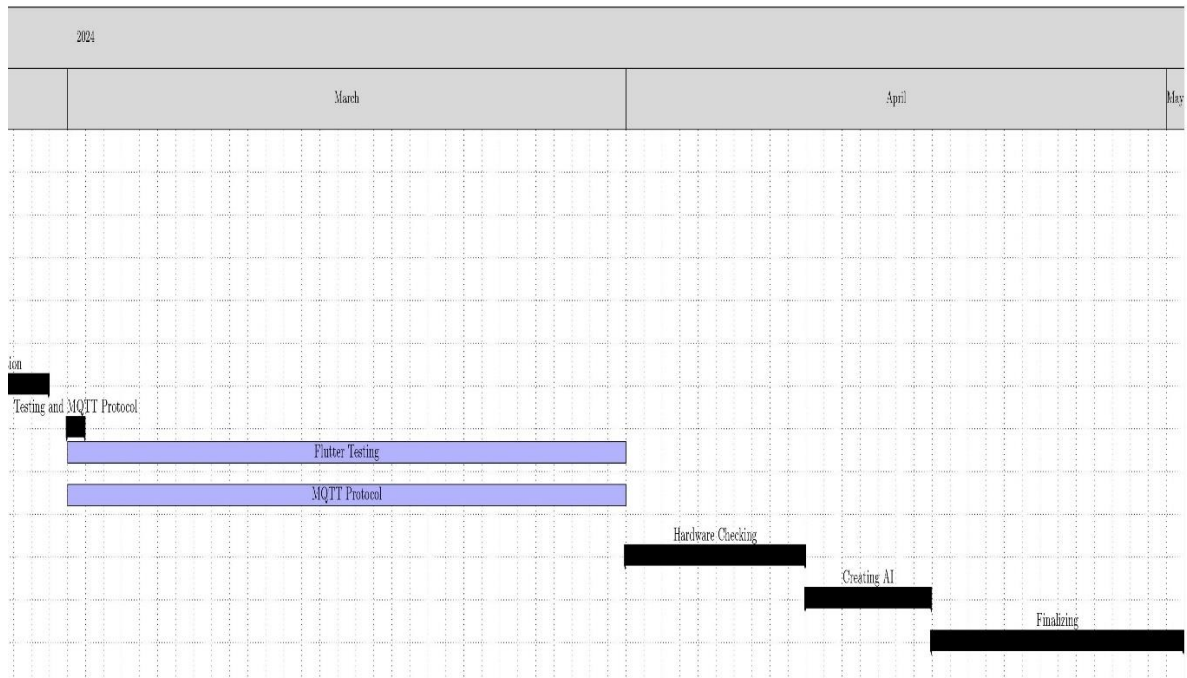
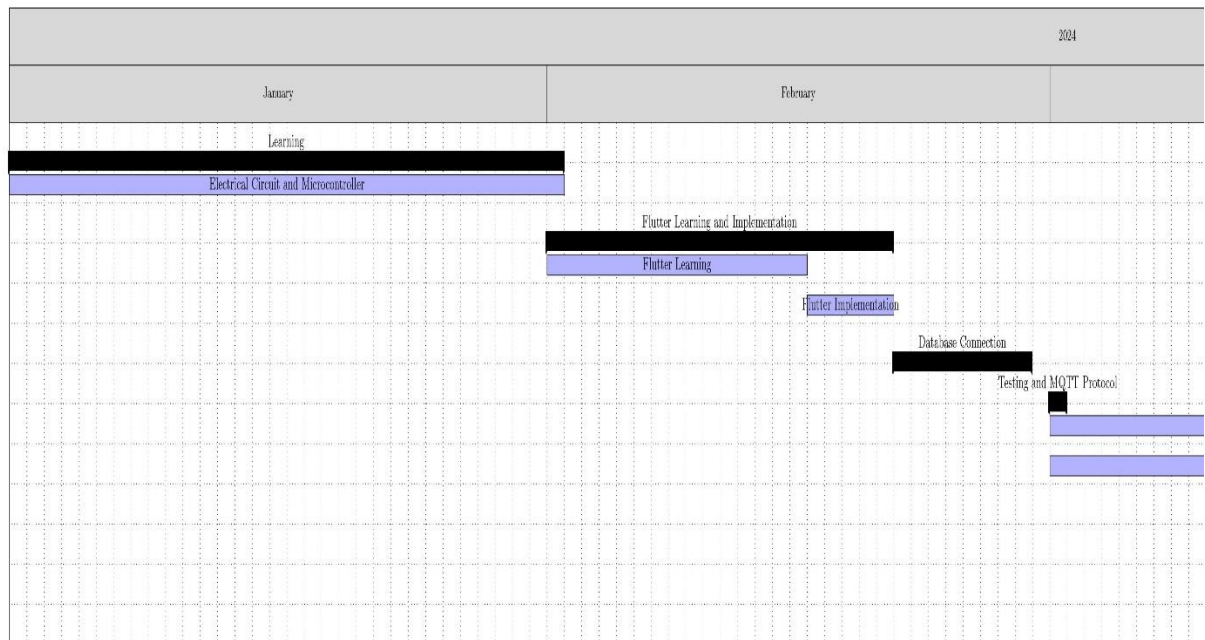
Components	Cost
Resistors	40
Transistor	45
Extra battery	80
Extra holder	20
9v Batter (x2)	50

Damaged while experimenting:

Components	Cost
2-relay (x2)	200
2 bulbs	120
Glue Gun	399
Glue Stick	30
Total cost	Rs.4239

Considering other things Approximately 5200-6000.

3.4 GANTT CHART



CHAPTER 4

ANALYSIS & DESIGN

4.1 PROPOSED METHODOLOGY

Understanding Requirements:

- Start by discussing and understanding the goals of the project with your team members.
- List down all the features you want your home automation system to have, keeping in mind the skills and resources available.

Selecting Components:

- Research and choose the hardware components needed for your project, such as microcontrollers (like ESP32 and Arduino Uno), sensors, and a webcam.
- Consider factors like cost, availability, and compatibility while making your selections.

Learning and Development:

- Begin learning about the selected hardware and software tools.
- Dive into tutorials and resources related to Flutter for building the mobile app, Firebase for authentication, and YOLOv8 for object detection.
- Start small by experimenting with individual components before integrating them into the larger system.

Setting Up Hardware:

- Once you have a good grasp of the components, start setting up the hardware.
- Follow tutorials and guides to properly connect and configure the microcontrollers, sensors, and webcam.

Coding and Integration:

- Begin coding the different components of your system.
- Develop the mobile app using Flutter, integrating features like remote control and real-time monitoring.

- Implement security measures using Firebase authentication and MQTT protocol for data exchange.
- Experiment with YOLOv8 for object detection and webcam functionality for remote surveillance.

Testing and Debugging:

- Test each component individually to ensure they work as expected.
- Integrate the components together and test the entire system.
- Debug any issues or errors that arise during testing.

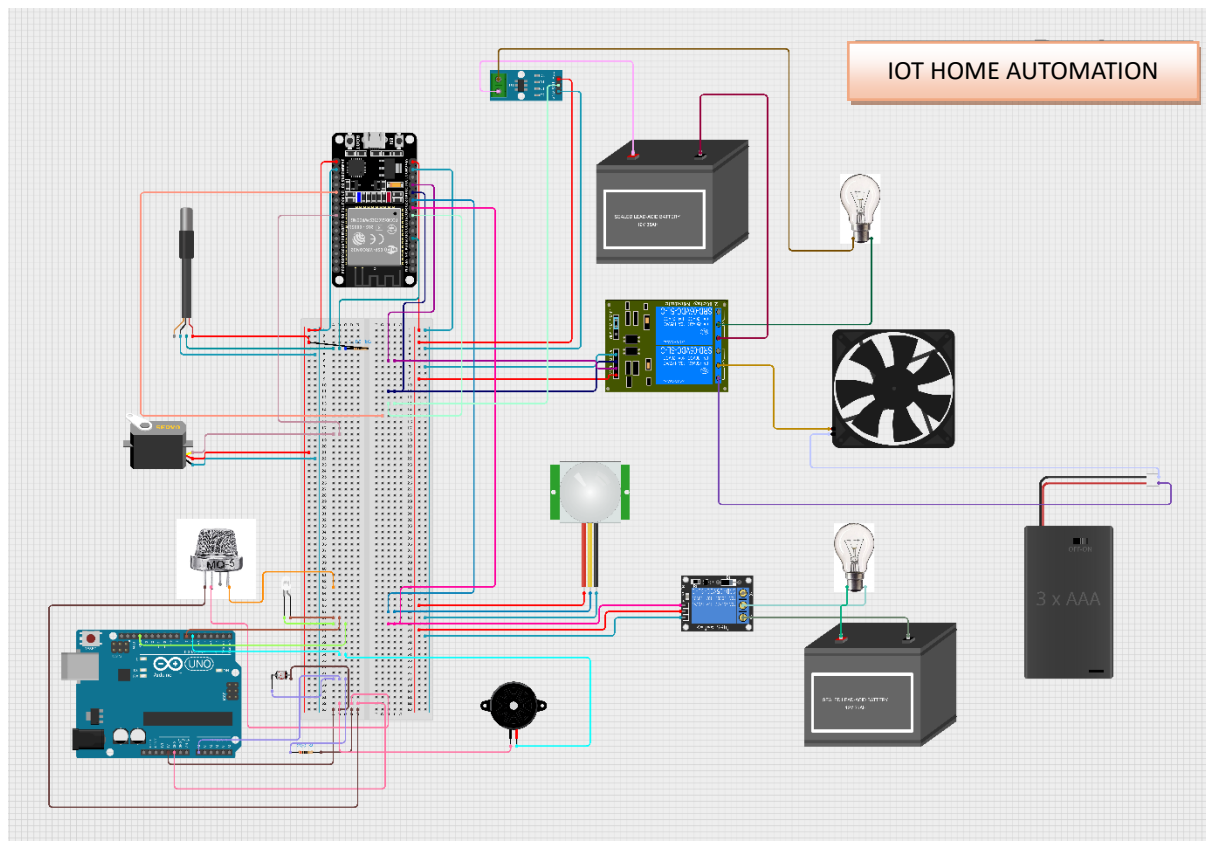
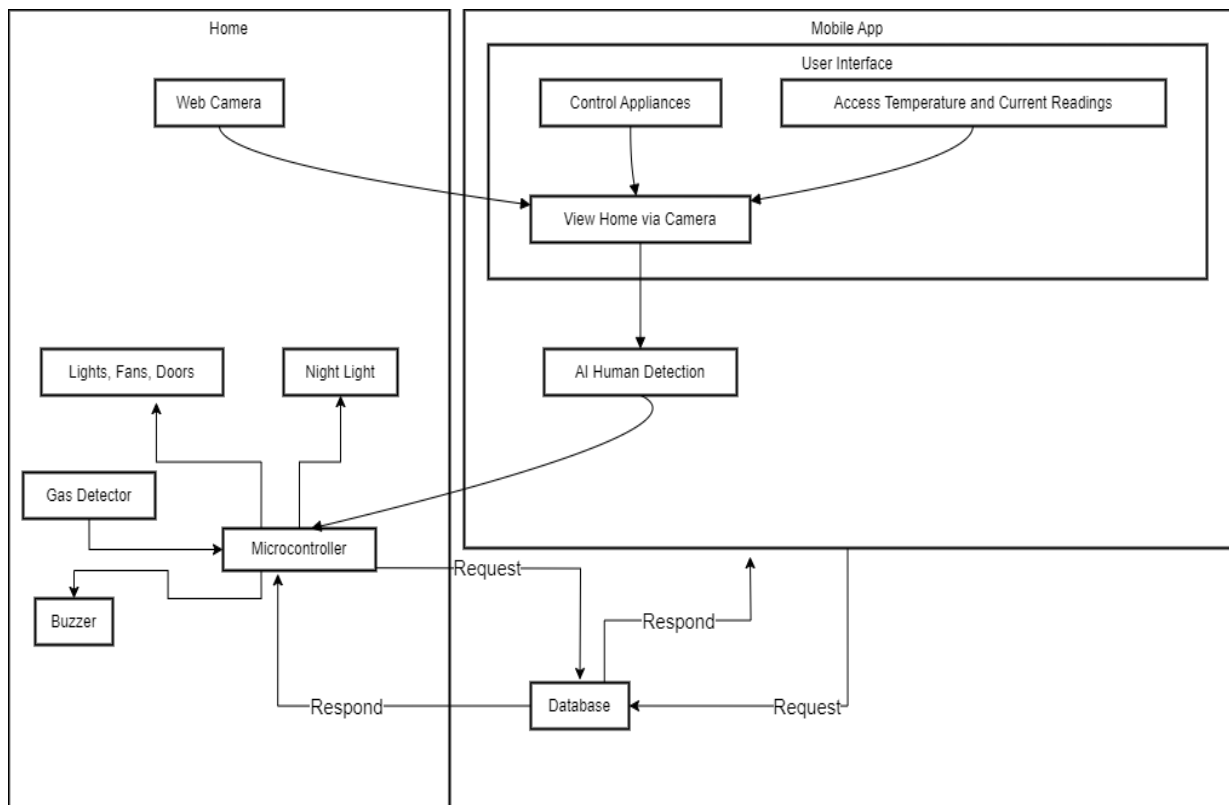
Documentation and Presentation:

- Document each step of your project, including setup instructions, code snippets, and troubleshooting tips.
- Prepare a presentation summarizing your project for your peers and instructors.

Feedback and Reflection:

- Seek feedback from your peers and instructors on your project.
- Reflect on the challenges you faced and what you learned throughout the process.
- Use feedback to improve your project and consider areas for future development.

4.2 SYSTEM ARCHITECTURE



4.3 MODULE DESCRIPTIONS

1. ESP32 Development Board:

- The ESP32 remains the core microcontroller board. It handles:
- Wi-Fi connectivity (or Wi-Fi Direct) for communication with the mobile app.
- Reading sensor data from connected sensors.
- Controlling actuators based on app commands.
- Processing data and communication using the chosen protocol.

2. Sensors:

- Temperature Sensor (DS18B20): Measures ambient temperature for functions like smart thermostat control or monitoring comfort levels.
- Motion Sensor (PIR sensor): Detects motion for security purposes (triggering alarms) or automated lighting control (turning on lights when someone enters a room).
- MQ-5 Gas Sensor: Detects the presence of various gases for air quality monitoring (important for potential gas leaks).
- Current sensor to get current readings.

3. Actuators:

- Servo Motor: Offers precise control over angular movement. It can be used for:
- Controlling blinds or curtains based on time of day or temperature.
- Adjusting the position of a camera for security monitoring.
- Buzzer to alter gas presence.
- Relay Module: Acts like a switch controlled by the ESP32. It can be used to turn on/off various appliances based on sensor data or user commands (e.g., lights, fans).

4. Communication Protocol (Wi-Fi Direct or MQTT):

- Establishes a connection between the ESP32 and the mobile app.
- MQTT (Message Queue Telemetry Transport): More flexible and scalable, using a central broker for message exchange but requiring additional configuration.

5. Flutter Mobile App:

- Developed using Flutter, the app provides a user interface for interacting with the smart home system. It will:
- Connect to the ESP32 via the chosen protocol (MQTT).
- Display sensor data (temperature, motion detection – if using motion sensor, air quality – if using MQ-5).
- Offer controls for actuators (e.g., turning on/off lights, adjusting fan speed, controlling servo motor position – if applicable).
- Potentially allow for setting schedules or automation rules for smart home devices.

6. Surveillance Module:

- This module enables remote surveillance of the home environment using a webcam.
- Yawcam software is utilized to capture live video feeds, which can be accessed through the mobile application.
- Object detection with YOLOv8 allows for automated surveillance and detection of human presence.

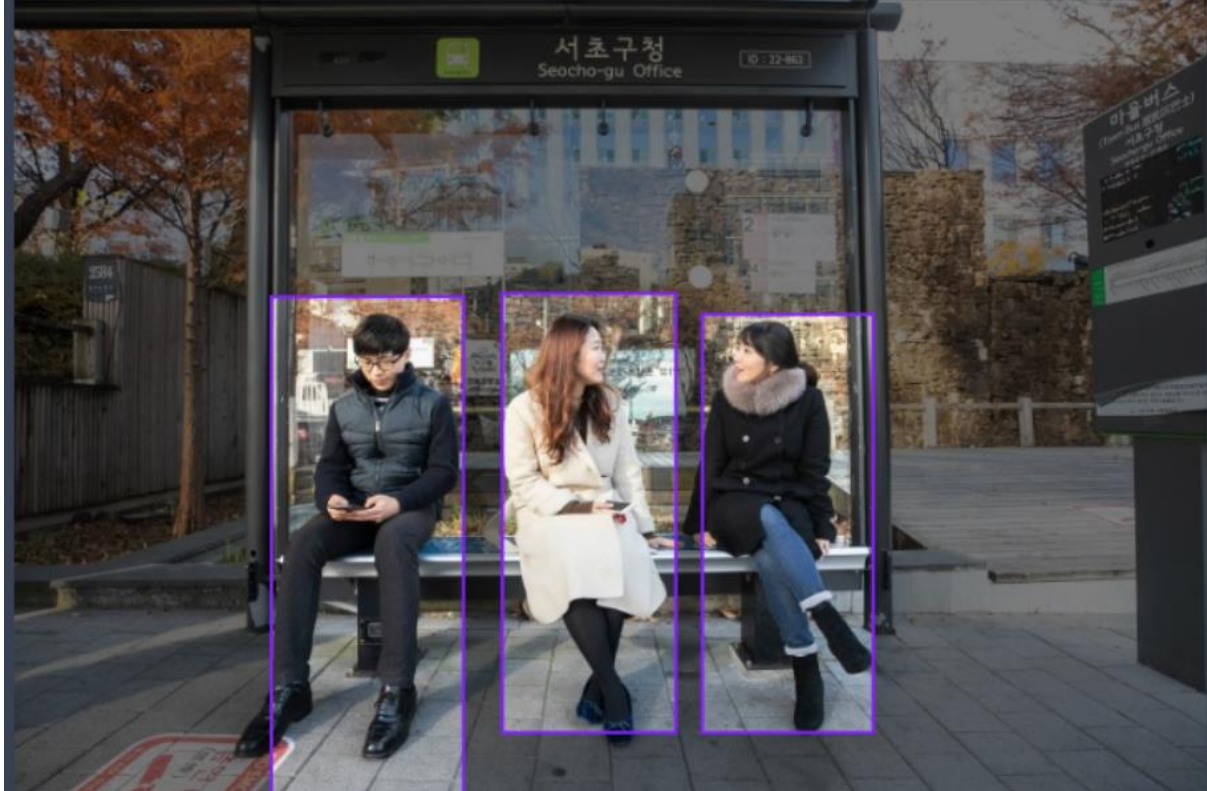
Additional Considerations:

- Breadboard and Jumper Wires: These help connect all the electronic components for prototyping and testing.
- Power Supply: Provides power to the ESP32 board and any additional components that require it (e.g., sensor modules, servo motor).

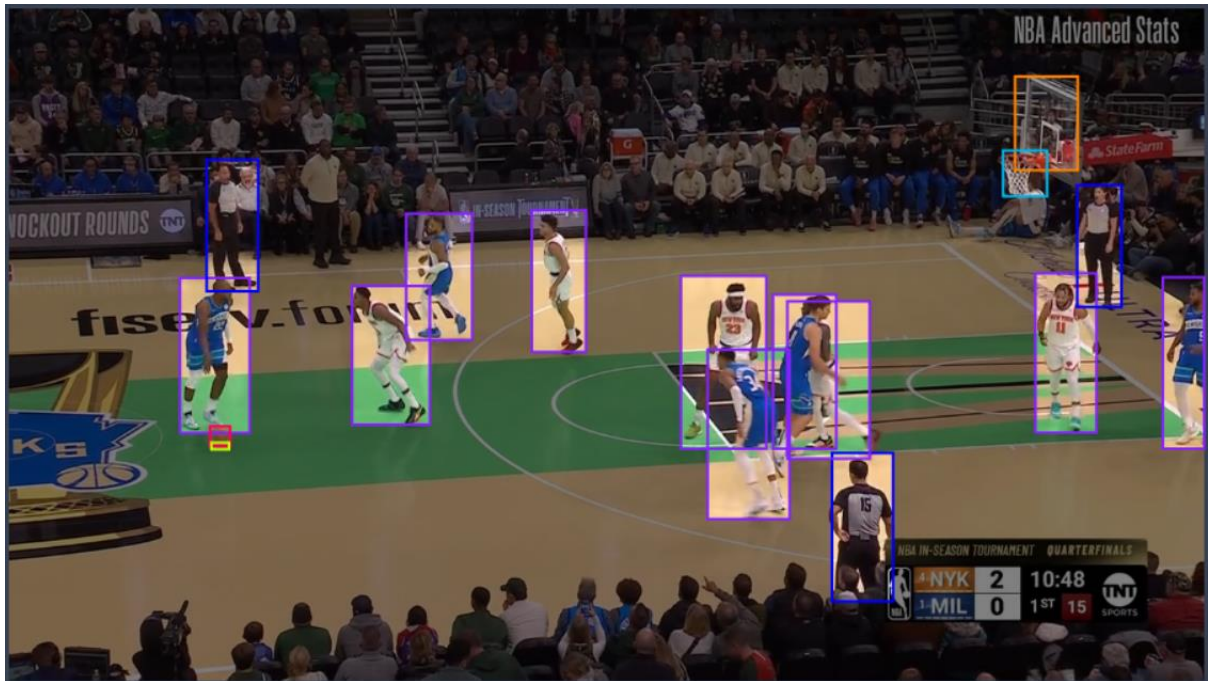
CHAPTER 5

IMPLEMENTATION & TESTING

5.1 DATASET







<https://universe.roboflow.com/>

5.2 SAMPLE CODE

Yolo v8:

```
from ultralytics import YOLO

import cv2
import math

import paho.mqtt.publish as mqtt

# start webcam

cap = cv2.VideoCapture(0)

cap.set(3, 1200)

cap.set(4, 720)


# model

model = YOLO("yolo-Weights/yolov8n.pt")


# object classes

classNames = ["person", "bicycle", "car", "motorbike", "aeroplane", "bus", "train", "truck", "boat",
              "traffic light", "fire hydrant", "stop sign", "parking meter", "bench", "bird", "cat",
              "dog", "horse", "sheep", "cow", "elephant", "bear", "zebra", "giraffe", "backpack", "umbrella",
              "handbag", "tie", "suitcase", "frisbee", "skis", "snowboard", "sports ball", "kite", "baseball
bat",
              "baseball glove", "skateboard", "surfboard", "tennis racket", "bottle", "wine glass", "cup",
              "fork", "knife", "spoon", "bowl", "banana", "apple", "sandwich", "orange", "broccoli",
              "carrot", "hot dog", "pizza", "donut", "cake", "chair", "sofa", "pottedplant", "bed",
              "diningtable", "toilet", "tvmonitor", "laptop", "mouse", "remote", "keyboard", "cell phone",
              "microwave", "oven", "toaster", "sink", "refrigerator", "book", "clock", "vase", "scissors",
              "teddy bear", "hair drier", "toothbrush"
]

while True:

    success, img = cap.read()

    results = model(img, stream=True)
```

```

# coordinates

for r in results:

    boxes = r.bboxes

    for box in boxes:

        # bounding box
        x1, y1, x2, y2 = box.xyxy[0]
        x1, y1, x2, y2 = int(x1), int(y1), int(x2), int(y2) # convert to int values

        # put box in cam
        cv2.rectangle(img, (x1, y1), (x2, y2), (255, 0, 255), 3)

        # confidence
        confidence = math.ceil((box.conf[0]*100))/100
        print("Confidence --->",confidence)

        # class name
        cls = int(box.cls[0])
        print("Class name -->", classNames[cls])

        if classNames[cls] == "person":
            mqtt.single("check", str("1"), hostname="91.121.93.94")
        else:
            mqtt.single("check", str("0"), hostname="91.121.93.94")

        # object details
        org = [x1, y1]
        font = cv2.FONT_HERSHEY_SIMPLEX
        fontScale = 1
        color = (255, 0, 0)

```

```
thickness = 2
```

```
cv2.putText(img, classNames[cls], org, font, fontScale, color, thickness)
```

```
cv2.imshow('Webcam', img)
```

```
if cv2.waitKey(1) == ord('q'):
```

```
    break
```

```
cap.release()
```

```
cv2.destroyAllWindows()
```

Flutter

Sign in page:

```
import 'package:firebase_auth/firebase_auth.dart';
```

```
import 'package:flutter/material.dart';
```

```
import 'package:mobileapplication/reusavle_widgets/reusable_widget.dart';
```

```
import 'package:mobileapplication/screens/home_screen.dart';
```

```
import 'package:mobileapplication/screens/reset_password.dart';
```

```
import 'package:mobileapplication/screens/signup_screen.dart';
```

```
import 'package:mobileapplication/utils/color_utils.dart';
```

```
class SignInScreen extends StatefulWidget {
```

```
  const SignInScreen({super.key});
```

```
  @override
```

```
  // ignore: library_private_types_in_public_api
```

```
  _SignInScreenState createState() => _SignInScreenState();
```

```
}
```

```
class _SignInScreenState extends State<SignInScreen> {
```

```
  final TextEditingController _passwordTextController = TextEditingController();
```

```

final TextEditingController _emailTextController = TextEditingController();

@override
Widget build(BuildContext context) {
  return Scaffold(
    body: Container(
      width: MediaQuery.of(context).size.width,
      height: MediaQuery.of(context).size.height,
      decoration: BoxDecoration(
        gradient: LinearGradient(colors: [
          hexStringToColor("#D16BA5"),
          hexStringToColor("#86A8E7"),
          hexStringToColor("#5FFBF1")
        ], begin: Alignment.topCenter, end: Alignment.bottomCenter)),
      child: SingleChildScrollView(
        child: Padding(
          padding: EdgeInsets.fromLTRB(
            20, MediaQuery.of(context).size.height * 0.2, 20, 0),
          child: Column(
            children: <Widget>[
              logoWidget("assets/Logo.png"),
              const SizedBox(
                height: 30,
              ),
              reusableTextField("Enter E-mail", Icons.person_outline, false,
                _emailTextController),
              const SizedBox(
                height: 20,
              ),
              reusableTextField("Enter Password", Icons.lock_outline, true,
                _passwordTextController),
              const SizedBox(

```

```

        height: 5,
      ),
      forgetPassword(context),
      firebaseUIButton(context, "Sign In", () {
        FirebaseAuth.instance
          .signInWithEmailAndPassword(
            email: _emailTextController.text,
            password: _passwordTextController.text)
          .then((value) {
            Navigator.push(context,
              MaterialPageRoute(builder: (context) => const HomeScreen()));
          }).onError((error, stackTrace) {
            print("Error ${error.toString()}");
          });
      }),
      signUpOption()
    ],
  ),
),
),
),
),
);
}

```

```

Row signUpOption() {
  return Row(
    mainAxisAlignment: MainAxisAlignment.center,
    children: [
      const Text("Don't have account?",
        style: TextStyle(color: Colors.white70)),
      GestureDetector(

```

```

onTap: () {
  Navigator.push(context,
    MaterialPageRoute(builder: (context) => const SignUpScreen()));
},
child: const Text(
  " Sign Up",
  style: TextStyle(color: Colors.white, fontWeight: FontWeight.bold),
),
),
],
);
}

```

```

Widget forgetPassword(BuildContext context) {
  return Container(
    width: MediaQuery.of(context).size.width,
    height: 35,
    alignment: Alignment.bottomRight,
    child: TextButton(
      child: const Text(
        "Forgot Password?",
        style: TextStyle(color: Colors.white70),
        textAlign: TextAlign.right,
      ),
      onPressed: () => Navigator.push(
        context, MaterialPageRoute(builder: (context) => const ResetPassword())),
    ),
  );
}

```

Sign- up page:

```

import 'package:firebase_auth/firebase_auth.dart';

import 'package:flutter/material.dart';

import 'package:mobileapplication/reusavle_widgets/reusable_widget.dart';

import 'package:mobileapplication/screens/home_screen.dart';

import 'package:mobileapplication/utils/color_utils.dart';

class SignUpScreen extends StatefulWidget {
  const SignUpScreen({super.key});

  @override
  // ignore: library_private_types_in_public_api
  _SignUpScreenState createState() => _SignUpScreenState();
}

class _SignUpScreenState extends State<SignUpScreen> {
  final TextEditingController _passwordTextController = TextEditingController();
  final TextEditingController _emailTextController = TextEditingController();
  final TextEditingController _userNameTextController = TextEditingController();

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      extendBodyBehindAppBar: true,
      appBar: AppBar(
        backgroundColor: Colors.transparent,
        elevation: 0,
        title: const Text(
          "Sign Up",
          style: TextStyle(fontSize: 24, fontWeight: FontWeight.bold),
        ),
      ),
      body: Container(

```

```

width: MediaQuery.of(context).size.width,
height: MediaQuery.of(context).size.height,
decoration: BoxDecoration(
  gradient: LinearGradient(colors: [
    hexStringToColor("#D16BA5"),
    hexStringToColor("#86A8E7"),
    hexStringToColor("#5FFBF1")
  ], begin: Alignment.topCenter, end: Alignment.bottomCenter)),
child: SingleChildScrollView(
  child: Padding(
    padding: const EdgeInsets.fromLTRB(20, 120, 20, 0),
    child: Column(
      children: <Widget>[
        const SizedBox(
          height: 20,
        ),
        reusableTextField("Enter UserName", Icons.person_outline, false,
          _userNameTextController),
        const SizedBox(
          height: 20,
        ),
        reusableTextField("Enter Email Id", Icons.person_outline, false,
          _emailTextController),
        const SizedBox(
          height: 20,
        ),
        reusableTextField("Enter Password", Icons.lock_outlined, true,
          _passwordTextController),
        const SizedBox(
          height: 20,
        ),
      ],
    ),
  ),
),

```

```

firebaseUIButton(context, "Sign Up", () {
  FirebaseAuth.instance
    .createUserWithEmailAndPassword(
      email: _emailTextController.text,
      password: _passwordTextController.text)
    .then((value) {
      print("Created New Account");
      Navigator.push(context,
        MaterialPageRoute(builder: (context) => const HomeScreen()));
    }).onError((error, stackTrace) {
      print("Error ${error.toString()}");
    });
})
],
),
))),
);
}
}

```

Homepage:

```

import 'package:firebase_auth/firebase_auth.dart';
import 'package:firebase_database/firebase_database.dart';
import 'package:flutter/material.dart';
import 'package:mobileapplication/screens/signin_screen.dart';
import 'package:mobileapplication/screens/webpage.dart';
import 'package:mobileapplication/screentest/door_control_page.dart';
import 'package:mobileapplication/temperature/temperaturescreen.dart';
import 'package:mobileapplication/utils/color_utils.dart';
//import 'package:mobileapplication/screentest/monitor_page.dart';

```

```

class HomeScreen extends StatefulWidget {
  const HomeScreen({super.key});

  @override
  _HomeScreenState createState() => _HomeScreenState();
}

class _HomeScreenState extends State<HomeScreen> {
  late DatabaseReference _devicesRef;

  int switch1 = 0;
  int switch2 = 0;
  int switch3 = 0;

  @override
  void initState() {
    super.initState();

    _devicesRef = FirebaseDatabase.instance.ref('devices');
    _devicesRef.onValue.listen((event) {
      final data = event.snapshot.value;
      if (data != null && data is Map<dynamic, dynamic>) {
        setState(() {
          switch1 = _getIntValueFromData(data, 'light1') ?? 0;
          switch2 = _getIntValueFromData(data, 'light2') ?? 0;
          switch3 = _getIntValueFromData(data, 'fan') ?? 0;
        });
      } else {
        // Handle the case when the data is null or not a Map
        print('Invalid data from Firebase: $data');
      }
    });
  }
}

```

```

int? _getIntValueFromData(Map<dynamic, dynamic> data, String key) {
  final value = data[key];
  if (value is int) {
    return value;
  } else if (value is String) {
    return int.tryParse(value);
  }
  return null;
}

```

```
@override
```

```

Widget build(BuildContext context) {
  return Scaffold(
    appBar: AppBar(
      title: const Text('Home Page'),
      actions: [
        IconButton(
          icon: const Icon(Icons.logout),
          onPressed: () {
            FirebaseAuth.instance.signOut().then((value) {
              print("Signed Out");
              Navigator.pushReplacement(
                context,
                MaterialPageRoute(
                  builder: (context) => const SignInScreen(),
                ),
              );
            });
          },
        ),
      ],
    ),
  );
}

```

```

],
leading: IconButton(
  icon: const Icon(Icons.arrow_back),
  onPressed: () {
    Navigator.of(context).pop();
  },
),
),
body: Container(
  decoration: BoxDecoration(
    gradient: LinearGradient(colors: [
      hexStringToColor("#D16BA5"),
      hexStringToColor("#86A8E7"),
      hexStringToColor("#5FFBF1")
    ],
  ),
)
),
child: Column(
  crossAxisAlignment: CrossAxisAlignment.center,
  mainAxisAlignment: MainAxisAlignment.center,
  children: [
    const Text(
      'Welcome to Home',
      style: TextStyle(
        fontSize: 24,
        fontWeight: FontWeight.bold,
        color: Colors.white,
      ),
    ),
    const SizedBox(height: 16),
    Expanded(

```

```

child: Column(
  mainAxisAlignment: MainAxisAlignment.center,
  children: [
    Row(
      mainAxisAlignment: MainAxisAlignment.spaceEvenly,
      children: [
        _buildButtonWithState(
          'assets/light.png',
          'Light 1',
          switch1,
          () {
            _updateSwitchState('light1', 1);
          },
          () {
            _updateSwitchState('light1', 0);
          },
        ),
        _buildButtonWithState(
          'assets/light.png',
          'Light 2',
          switch2,
          () {
            _updateSwitchState('light2', 1);
          },
          () {
            _updateSwitchState('light2', 0);
          },
        ),
      ],
    ),
    const SizedBox(height: 16),

```

```

Row(
  mainAxisAlignment: MainAxisAlignment.spaceEvenly,
  children: [
    _buildButtonWithState(
      'assets/fan.png',
      'Fan',
      switch3,
      () {
        _updateSwitchState('fan', 1);
      },
      () {
        _updateSwitchState('fan', 0);
      },
    ),
    _buildRedirectButton('assets/door.png', 'Door', () {
      Navigator.push(
        context,
        MaterialPageRoute(
          builder: (context) => const DoorControlPage());
    })),
  ],
),
const SizedBox(height: 16),
Row(
  mainAxisAlignment: MainAxisAlignment.spaceEvenly,
  children: [
    _buildRedirectButton('assets/monitor.png', 'Readings', () {
      Navigator.push(
        context,
        MaterialPageRoute(
          builder: (context) => const SensorPage());
    })),
  ],
),

```

```

    )),
    _buildRedirectButton('assets/cctv.png', 'Monitor', () {
      Navigator.push(
        context,
        MaterialPageRoute(
          builder: (context) => const WebsitePage(url: 'http://127.0.0.1:8081'),
        ),
      );

      // Navigate to temperature monitor page
    )),
  ],
),
],
),
),
],
),
),
];
);
}

```

```

Widget _buildButtonWithState(String imagePath, String label, int state,
  VoidCallback onPressedOn, VoidCallback onPressedOff) {
  return Column(
    children: [
      Material(
        color: Colors.transparent,
        child: InkWell(
          onTap: state == 0 ? onPressedOn : onPressedOff,
          child: Padding(

```

```

padding: const EdgeInsets.all(8.0),
child: Image.asset(
  imagePath,
  width: 64,
  height: 64,
),
),
),
),
const SizedBox(height: 6),
Text(
  label,
  style: const TextStyle(fontSize: 16),
),
],
);
}

```

```

Widget _buildRedirectButton(
  String imagePath, String label, VoidCallback onPressed) {
return Column(
  children: [
    Material(
      color: Colors.transparent,
      child: InkWell(
        onTap: onPressed,
        child: Padding(
          padding: const EdgeInsets.all(8.0),
          child: Image.asset(
            imagePath,
            width: 64,

```

```

        height: 64,
    ),
    ),
    ),
    ),
    const SizedBox(height: 6),
    Text(
        label,
        style: const TextStyle(fontSize: 16),
    ),
  ],
);
}

void _updateSwitchState(String device, int value) {
  _devicesRef.child(device).set(value);
}
}

```

Arduino:

```

#include <WiFi.h>

#include <ESP32Firebase.h>

#include <PubSubClient.h>

#include <OneWire.h>

#include <DallasTemperature.h>

#include <ESP32Servo.h>

#define _SSID "Poda eruma"    // Your WiFi SSID

#define _PASSWORD "VSweta0078" // Your WiFi Password

```

```

#define mqtt_server "91.121.93.94"

#define REFERENCE_URL "https://iot-app-f5138-default-rtdb.asia-
southeast1.firebaseio.com/" // Your Firebase project reference url

// #define API_KEY "AlzaSyDzL_-th-PXinHPlp6PapqB8cq_MTWTS04"

Firebase firebase(REFERENCE_URL);

#define topic "check"

WiFiClient espClient;
PubSubClient client(espClient);

Servo myServo;
#define ONE_WIRE_BUS 14

const int motion = 33;
const int light1_relay = 15;
const int light2_relay = 21;
const int fan_relay = 5;
const int servo = 18;

#define VIN 25

const float VCC = 5.0;
const int model = 1;

float cutOffLimit = 1.01;

float sensitivity[] = {
    0.185,
    0.100,
    0.066

};

```

```
const float QOV = 0.5 * VCC;// set quiescent Output voltage of 0.5V  
float voltage;// internal variable for voltage
```

```
OneWire oneWire(ONE_WIRE_BUS);  
DallasTemperature sensors(&oneWire);
```

```
void setup(){  
  Serial.begin(115200);  
  pinMode(light1_relay, OUTPUT);  
  pinMode(light2_relay, OUTPUT);  
  pinMode(fan_relay, OUTPUT);  
  pinMode(motion, INPUT);
```

```
  myServo.attach(servo);
```

```
  //connect wifi  
  WiFi.mode(WIFI_STA);  
  WiFi.disconnect();  
  delay(1000);
```

```
  // Connect to WiFi  
  Serial.println();  
  Serial.println();  
  Serial.print("Connecting to: ");  
  Serial.println(_SSID);  
  WiFi.begin(_SSID, _PASSWORD);
```

```
  while (WiFi.status() != WL_CONNECTED) {  
    delay(500);  
    Serial.print("-");
```

```

}

Serial.println("");
Serial.println("WiFi Connected");

// Print the IP address
Serial.print("IP Address: ");
Serial.print("http://");
Serial.print(WiFi.localIP());
Serial.println("/");

client.setServer(mqtt_server,1883);

digitalWrite(light1_relay, HIGH);
digitalWrite(light2_relay, HIGH);
digitalWrite(fan_relay, HIGH);
}

void loop() {

if (firebase.getInt("devices/door") == 1) {
  Serial.print(firebase.getInt("devices/door"));
  openDoor();
} else if (firebase.getInt("devices/door") == 2) {
  closeDoor();
}

if(firebase.getInt("devices/light1") == 1)//Firabase: light1 on
{

```

```

digitalWrite(light1_relay,LOW);

}else if(firebase.getInt("devices/light1") == 0){
    digitalWrite(light1_relay, HIGH);

}

if(firebase.getInt("devices/light2") == 1)//Firabase: light2 on
{
    digitalWrite(light2_relay,LOW);

}else if(firebase.getInt("devices/light2") == 0){
    digitalWrite(light2_relay, HIGH);

}

if(firebase.getInt("devices/fan") == 1 )//Firabase: fan on
{
    digitalWrite(fan_relay,LOW);

}else if(firebase.getInt("devices/fan") == 0){
    digitalWrite(fan_relay,HIGH);

}

int motionState = digitalRead(motion);
if(motionState == HIGH){
    firebase.setInt("devices/light1",1);
    firebase.setInt("devices/fan",1);
    digitalWrite(light1_relay, LOW);
    digitalWrite(fan_relay, LOW);
}

```

```

    readTemperature();
    readcurrent();

}

void openDoor() {
    myServo.write(180);
}

void closeDoor() {
    myServo.write(60);
}

void readTemperature(){
    sensors.requestTemperatures();
    float temperatureC = sensors.getTempCByIndex(0);
    if (temperatureC != DEVICE_DISCONNECTED_C) {
        Serial.print("Temperature: ");
        Serial.print(temperatureC);
        Serial.println(" °C");

        firebase.setFloat("sensors/temperature",temperatureC);
    } else {
        Serial.println("Error: Could not read temperature data");
        Serial.print(temperatureC);
    }
}

void readcurrent(){
    float voltage_raw = (5.0 / 1023.0)* analogRead(VIN);// Read the voltage from sensor

```

```
voltage = voltage_raw - QOV + 0.012 ;// 0.000 is a value to make voltage zero when there is no current
```

```
float current = voltage / 1;
```

```
if(abs(current) > cutOffLimit ){
```

```
    Serial.print("V: ");
```

```
    Serial.print(voltage,3);// print voltage with 3 decimal places
```

```
    Serial.print("V, I: ");
```

```
    Serial.print(current,2); // print the current with 2 decimal places
```

```
    Serial.println("A");
```

```
    firebase.setFloat("sensors/current",voltage);
```

```
}else{
```

```
    Serial.println("No Current");
```

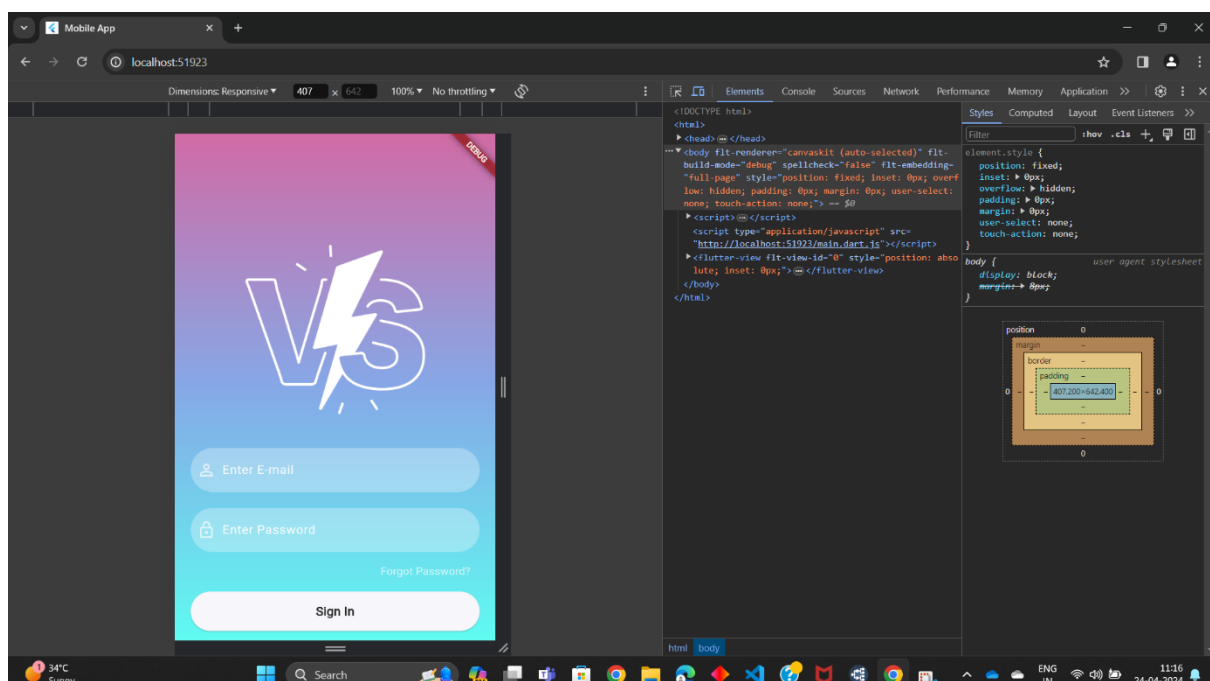
```
    firebase.setFloat("sensors/current",0.00);
```

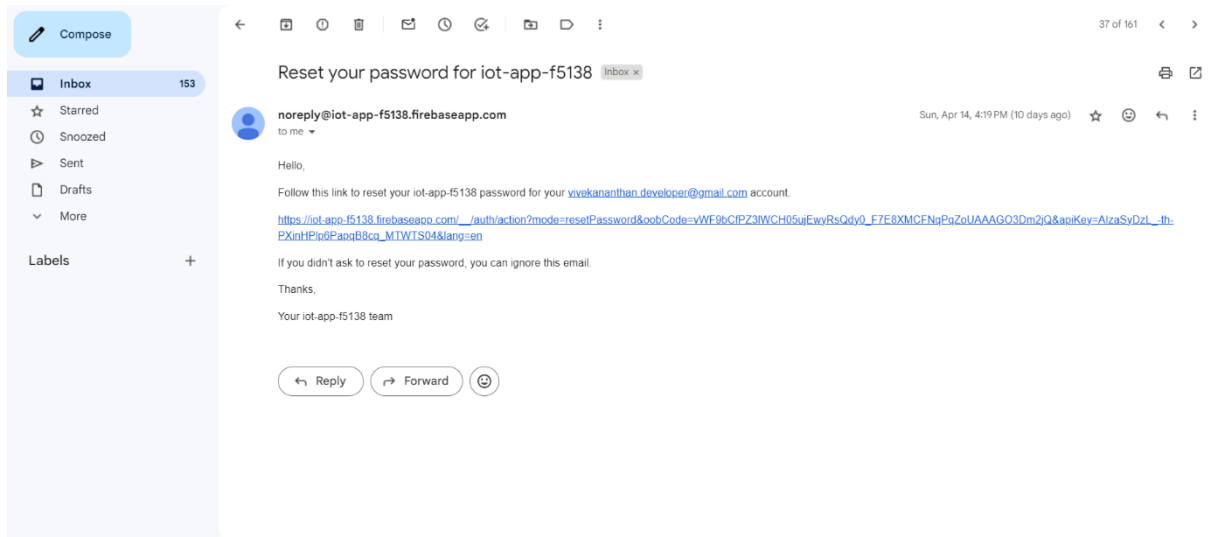
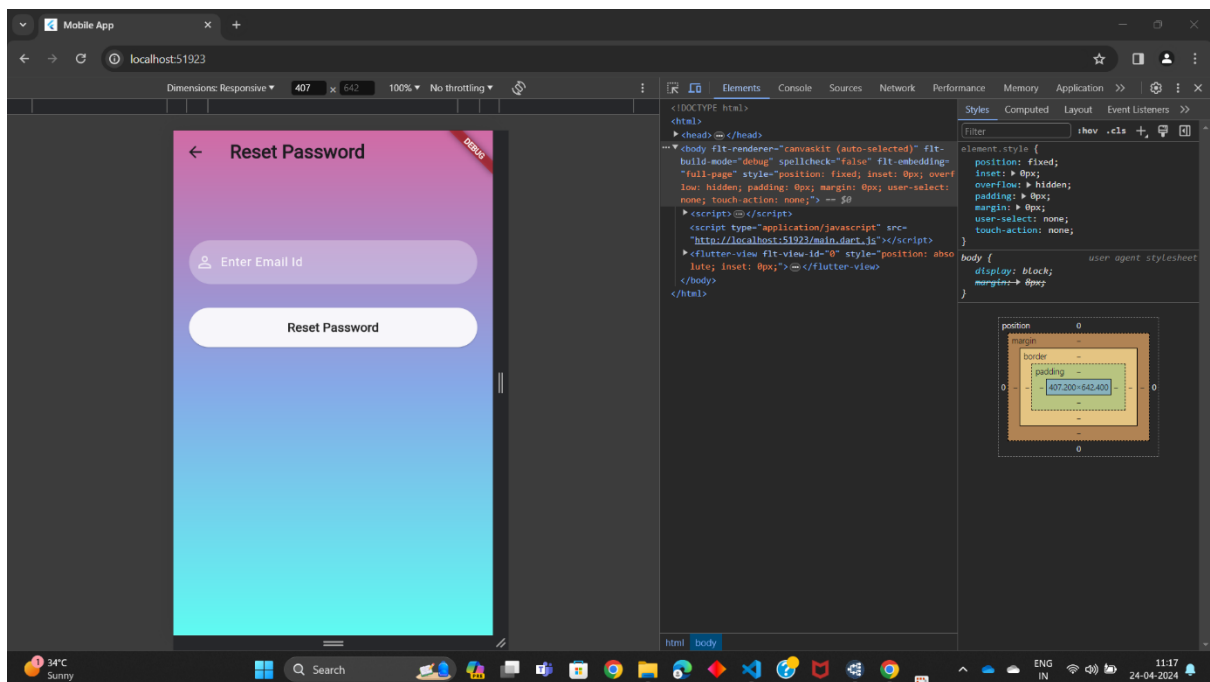
```
}
```

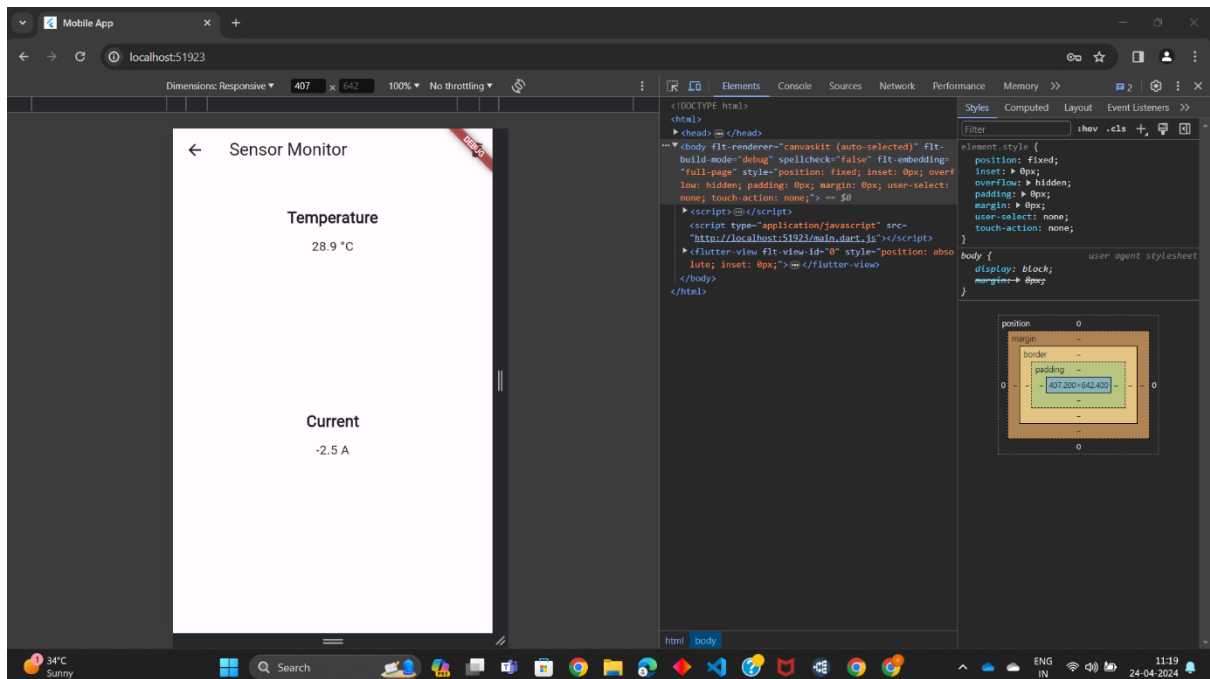
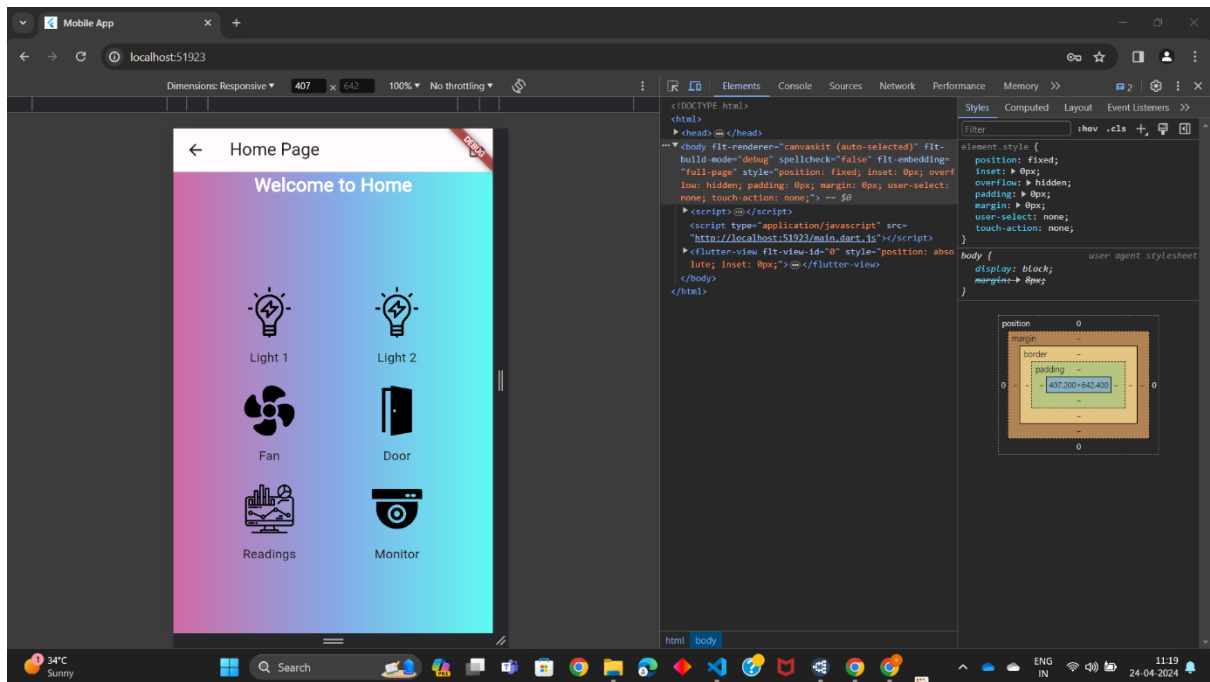
```
delay(500);
```

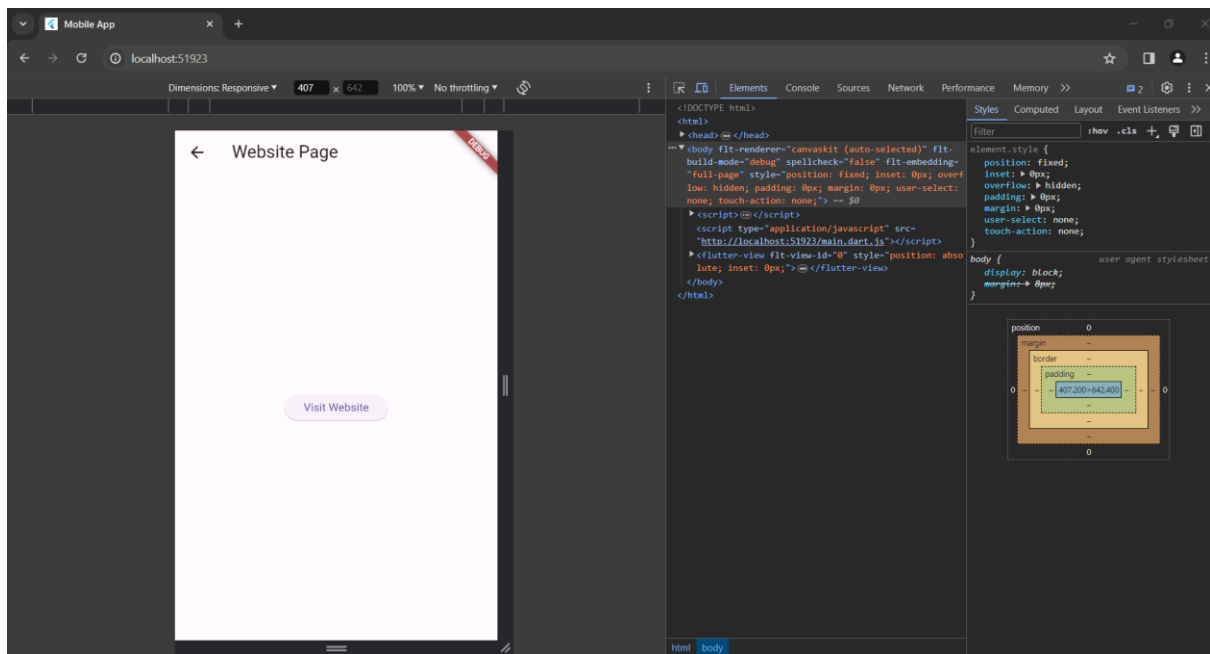
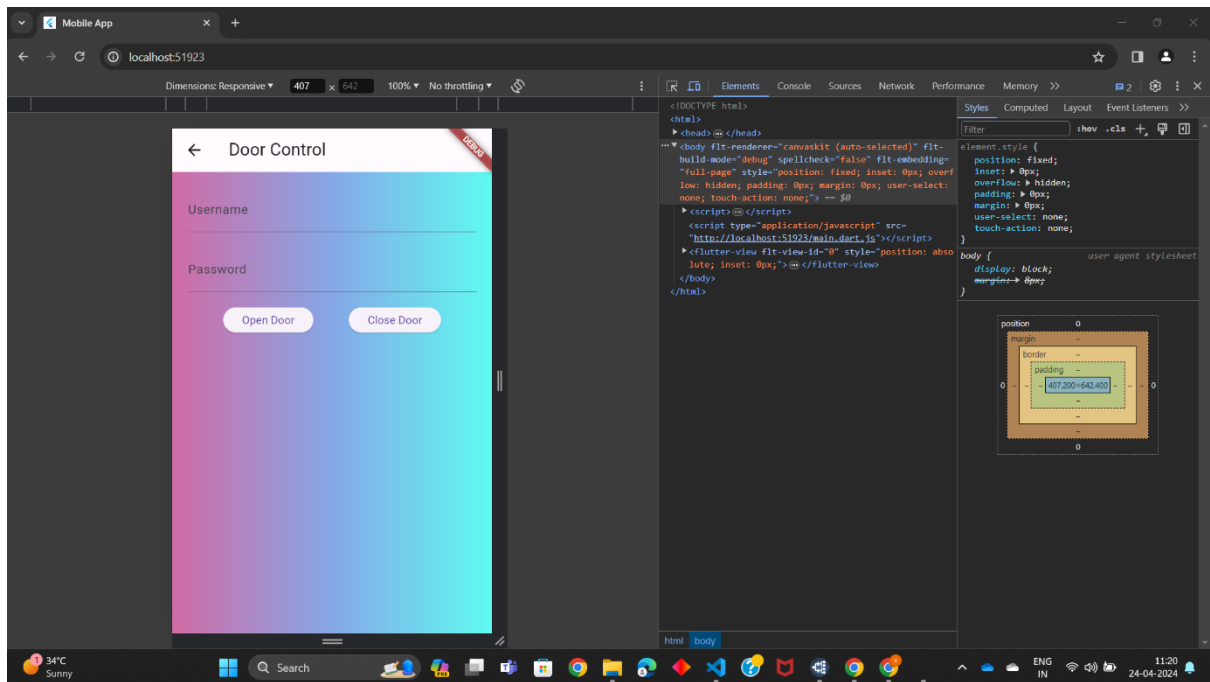
```
}
```

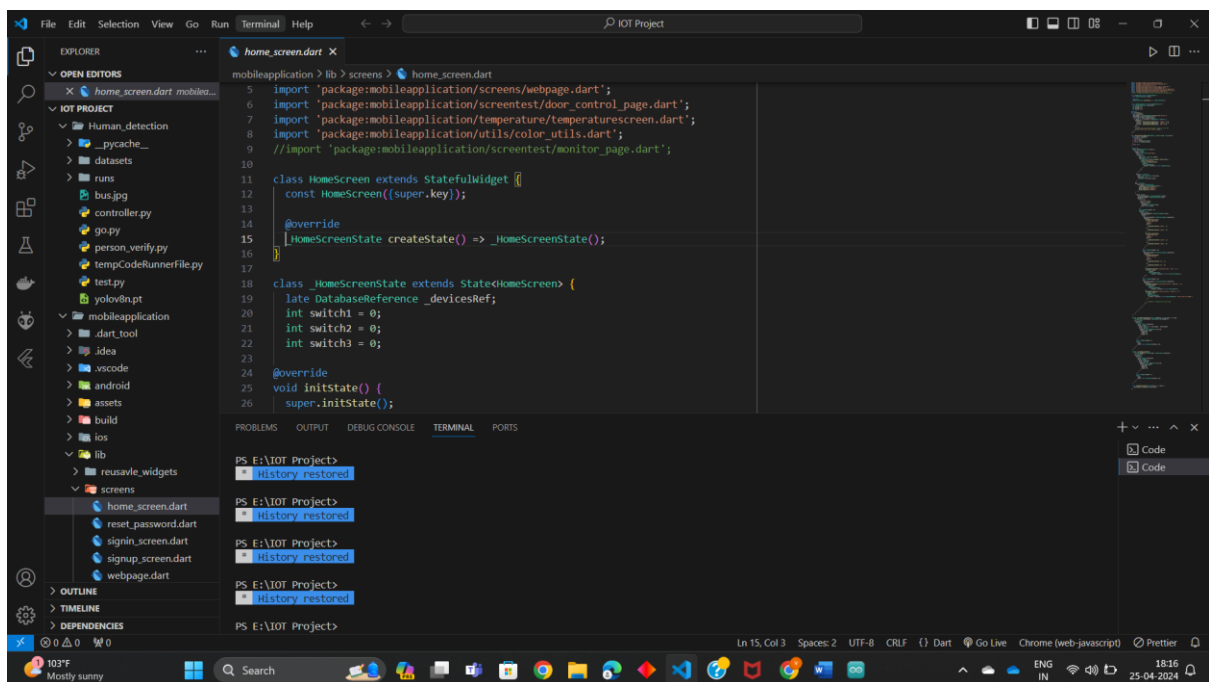
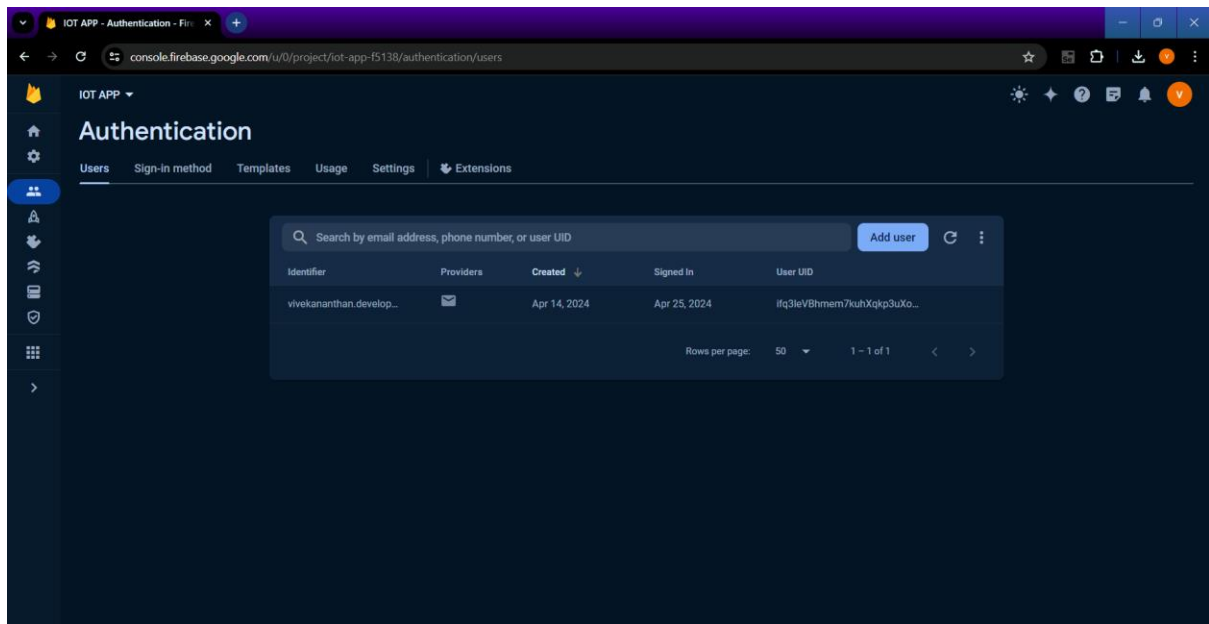
5.3 SAMPLE OUTPUT

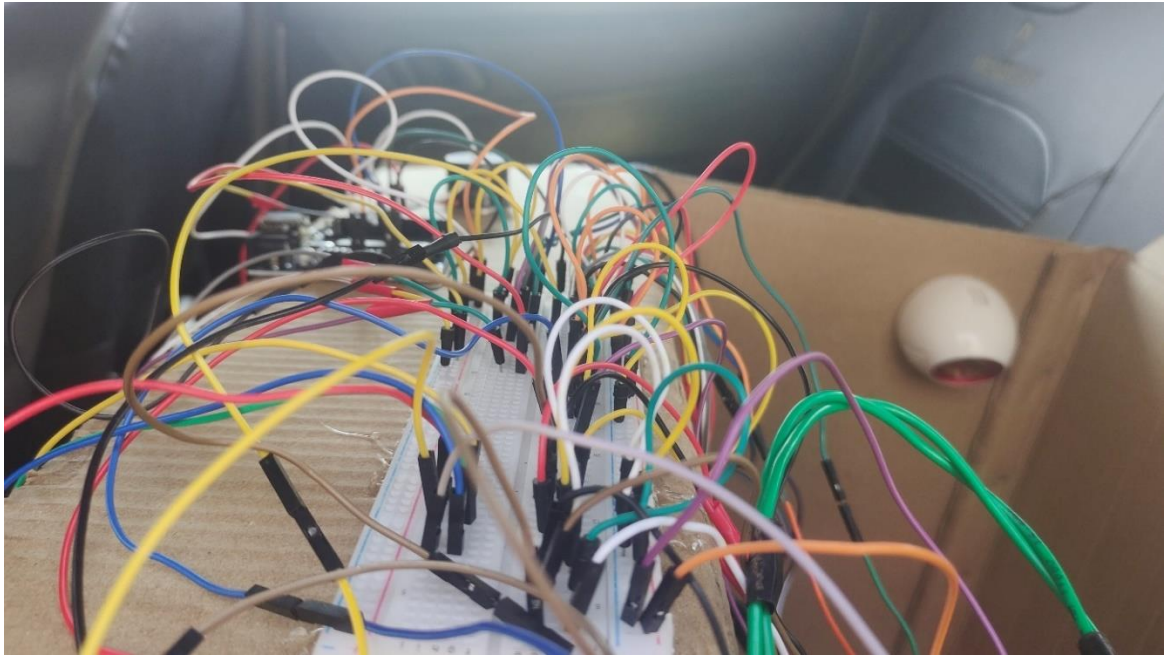


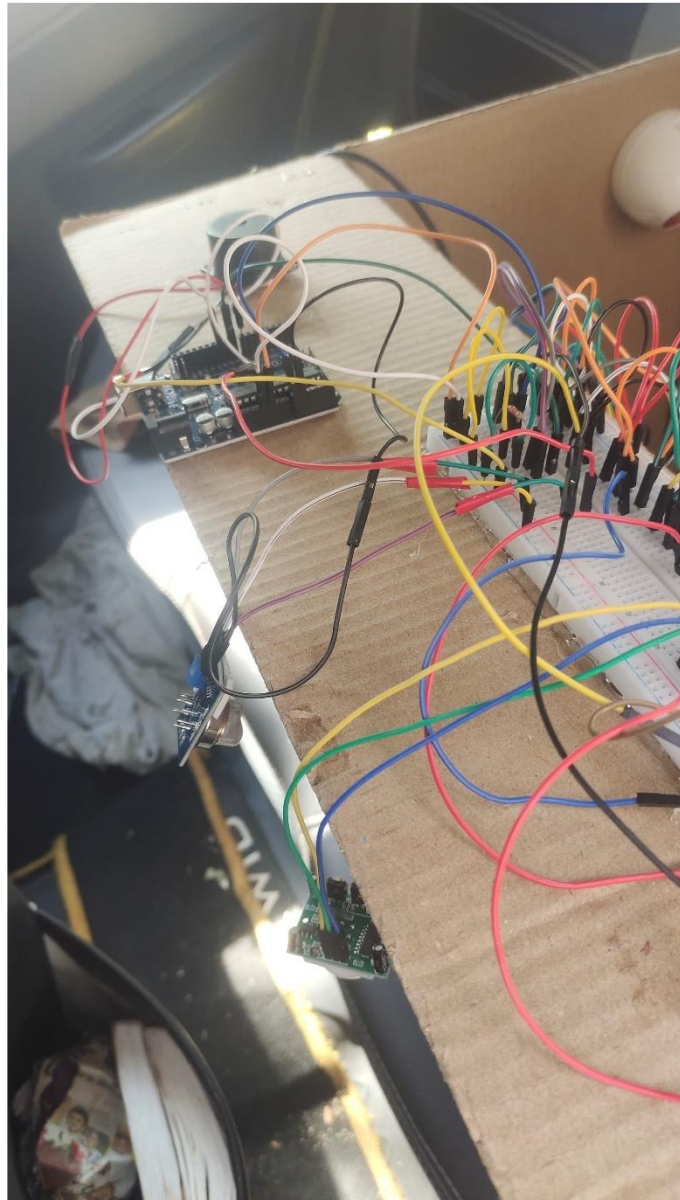


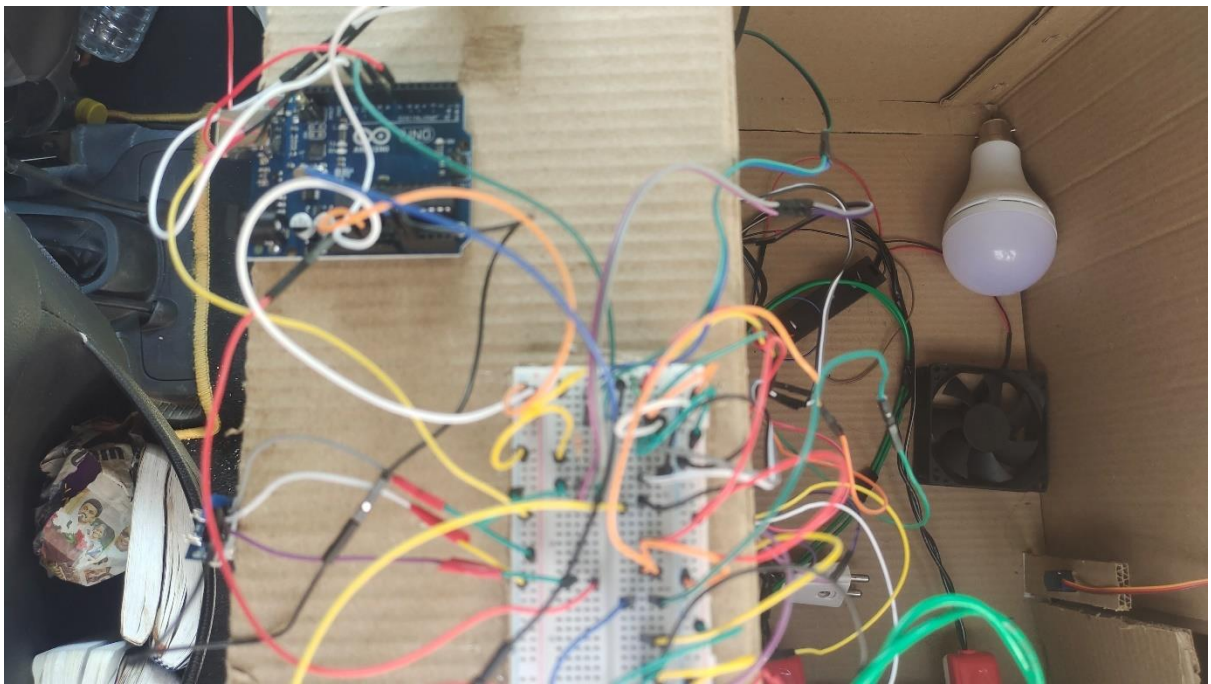
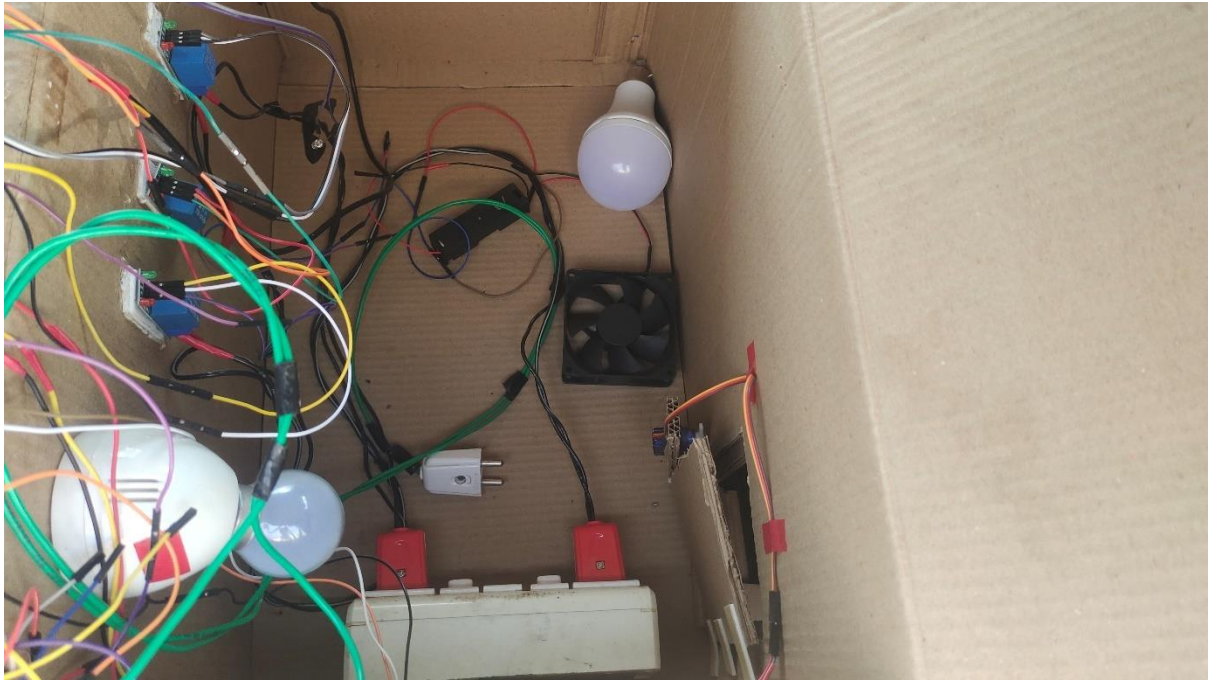


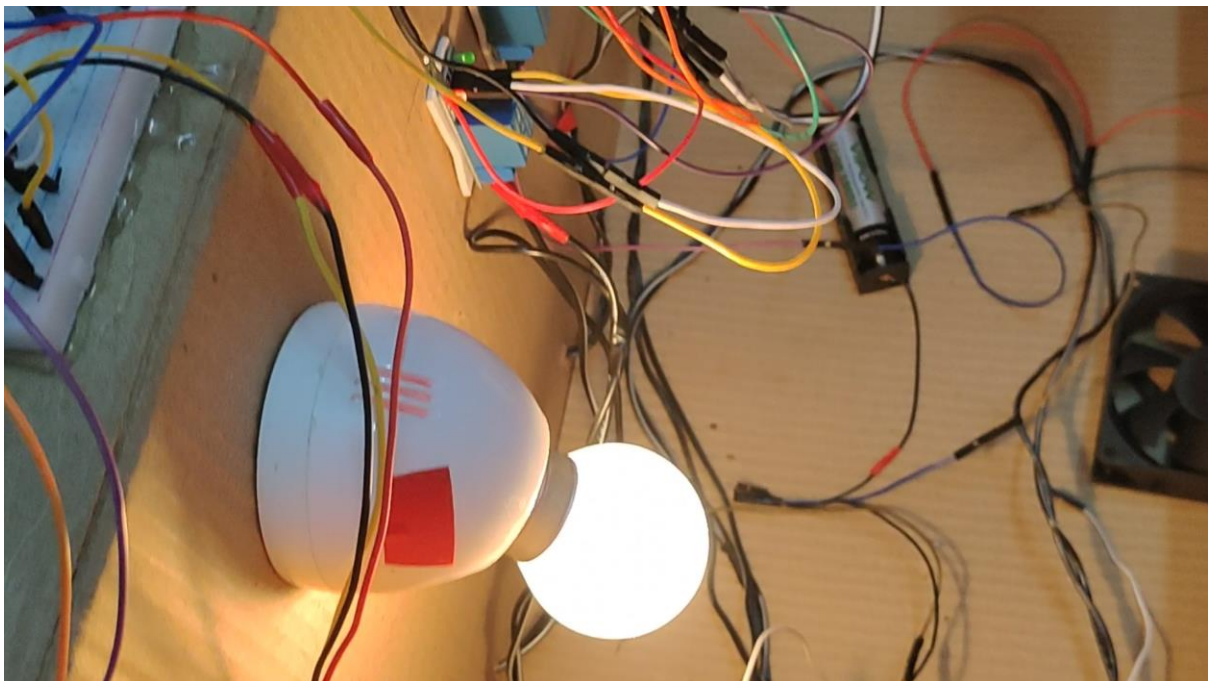
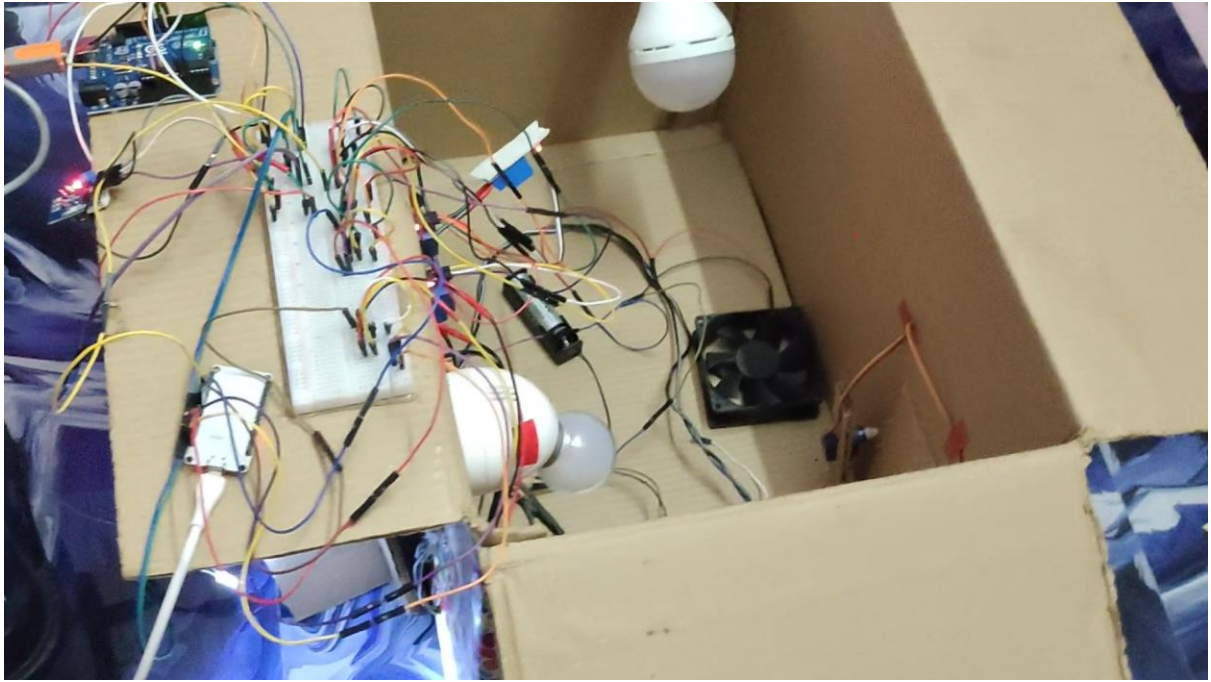












5.4 TEST PLAN & DATA VERIFICATION

HTC - Hardware Test case

Test case no	Sensor Testing	Expected Output	Actual Output	Result
TC_1	Temperature in fahrenheit	66- 78 F	76 F	Pass
TC_2	Smoke sensor	LPG gas detected	LPG gas detected	Pass
TC_3	Servo motor	90 degrees rotated	90 degrees rotated	Pass
TC_4	Relay	On	ON	Pass
TC_5	Blink Led	HIGH: ON	HIGH: ON	Pass
TC_6	Fan	Spinning	Spinning	Pass
TC_7	WIFI connection	WIFI Connected	WIFI Not Connected	Fail
TC_8	WIFI Connection	WIFI Connected	WIFI Connected	Pass
TC_9	Motion detection	HIGH	LOW	Fail
TC_10	Motion detection	HIGH	HIGH	Pass

STC – Software test case

Test case no	Input	Expected Output	Actual Output	Result
TC_1	Correct Username and password	Home page	Home page	Pass
TC_2	Incorrect Username and password	Incorrect username/password	Incorrect username/password	Pass
TC_3	Send 1 to database	Updated in database	Updated in database	Pass
TC_4	Get temperature value	Temperature value	35.5	Pass
TC_5	Detect Human	Person 1	Error	Fail
TC_6	Detect Human_2	Person 1	Not Updated	Fail
TC_7	Detect Human_3	Person 1	Person	Pass
TC_8	Mqtt connection	Connected	Connected	Pass
TC_9	Reset password	Send link to mail	Send link to mail	Pass
TC_10	Live webcam	Redirect to website	Redirect to website	Pass

CHAPTER 6

RESULTS

6.1 RESEARCH FINDINGS

IoT Technologies and Protocols:

Research on different IoT technologies, communication protocols (e.g., MQTT, HTTP), and standards (e.g., Zigbee, Z-Wave) to understand their suitability for your project requirements.

Microcontrollers and Sensors:

Investigate different microcontroller platforms (e.g., ESP32, Arduino Uno) and sensors (e.g., temperature, gas, light sensors) to select the most appropriate ones for your project based on factors such as functionality, cost, and ease of integration.

Mobile Application Development:

Explore frameworks and tools for developing mobile applications, such as Flutter, React Native, or native development for iOS and Android platforms. Evaluate their features, development workflow, and compatibility with your project requirements.

Security Measures:

Research best practices for implementing security measures in IoT systems, including authentication mechanisms, encryption techniques, and secure data transmission protocols (e.g., HTTPS, SSL/TLS).

Object Detection and Computer Vision:

Investigate object detection algorithms and frameworks like YOLO (You Only Look Once) and OpenCV for detecting human presence in the room using webcam feeds. Understand their capabilities, performance, and integration with your system.

Cloud Services:

Explore cloud platforms like Firebase, AWS IoT, or Azure IoT for backend services such as user authentication, real-time data storage, and remote device management. Evaluate their features, pricing, and scalability for your project.

Data Visualization and Analytics:

Research tools and libraries for visualizing sensor data and analytics to provide insights into home environment metrics such as temperature, humidity, and energy usage. Consider options like Grafana, Plotly, or custom dashboards built with web technologies.

User Experience Design:

Study principles of user experience (UX) design to create an intuitive and user-friendly interface for controlling and monitoring the home automation system through the mobile application. Consider aspects such as navigation, layout, and feedback mechanisms.

6.2 RESULT ANALYSIS & EVALUATION METRICS

Functionality Testing:

- **Metric:** Accuracy of device control and automation.
- **Evaluation:** Test the accuracy and responsiveness of device control commands sent from the mobile application to the hardware components. Measure the success rate of automated actions triggered by environmental factors such as human presence detection or gas detection.

Security Evaluation:

- **Metric:** Security robustness and resilience.
- **Evaluation:** Assess the effectiveness of security measures implemented, including username/password authentication, data encryption, and secure communication protocols. Conduct penetration testing to identify vulnerabilities and ensure resistance against unauthorized access or malicious attacks.

System Stability and Reliability:

- **Metric:** System uptime and reliability.

- **Evaluation:** Monitor the system's stability over an extended period to identify any issues such as crashes, freezes, or failures. Measure uptime and availability to ensure consistent performance under varying conditions, including network disruptions or hardware failures.

User Experience (UX) Evaluation:

- **Metric:** User satisfaction and ease of use.
- **Evaluation:** Gather user feedback through surveys, interviews, or usability testing sessions to assess the overall satisfaction with the mobile application interface and the home automation system's usability. Measure user engagement, task completion rates, and error rates to identify areas for improvement in the user experience.

Performance of Object Detection:

- **Metric:** Accuracy and speed of object detection.
- **Evaluation:** Evaluate the accuracy of object detection algorithms (e.g., YOLOv8) in detecting human presence in the room based on webcam feeds. Measure metrics such as precision, recall, and processing speed to assess the algorithm's performance in real-time scenarios.

Energy Efficiency:

- **Metric:** Energy consumption and optimization.
- **Evaluation:** Measure the energy consumption of the home automation system components (e.g., microcontrollers, sensors) under different operating conditions. Assess the system's efficiency in optimizing energy usage, such as automatically turning off lights and fans when not needed, to promote energy conservation.

Data Transmission Efficiency:

- **Metric:** Data transmission latency and reliability.
- **Evaluation:** Analyse the latency and reliability of data transmission between the mobile application, hardware components, and cloud services. Measure metrics such as message delivery time, packet loss rate, and communication overhead to ensure efficient and reliable data exchange.

CONCLUSION

In conclusion, the development and implementation of the IoT-based home automation system presented in this project have demonstrated significant strides towards addressing the modern-day needs of convenience, security, and energy efficiency in residential environments. Through meticulous research, careful planning, and iterative development, a comprehensive solution has been crafted that seamlessly integrates various hardware and software components to create a user-friendly and efficient system.

The project's success lies in its ability to fulfil the objectives set forth, including remote control of home devices through a mobile application, real-time monitoring of environmental metrics, robust security measures, and intelligent automation based on object detection. By leveraging technologies such as Flutter for mobile application development, Firebase for authentication and data exchange, and YOLOv8 for object detection, the system offers a sophisticated yet accessible platform for users to manage their homes remotely.

Furthermore, the project has not only delivered a functional home automation system but also provided valuable learning opportunities for students and enthusiasts interested in IoT technology. Through hands-on experimentation, research findings, and result analysis, participants have gained insights into various aspects of IoT development, including hardware integration, software development, security implementation, and user experience design.

Looking ahead, there are several avenues for future exploration and improvement, including enhancing system scalability, optimizing energy efficiency, and expanding the range of supported devices and functionalities. Additionally, continuous monitoring, feedback collection, and iterative refinement will be essential to ensure the system's reliability, usability, and relevance in the ever-evolving landscape of home automation technology.

In summary, this project serves as a testament to the transformative potential of IoT technology in revolutionizing the way we interact with our living spaces. By combining innovation with practicality and education, we pave the way for a smarter, safer, and more sustainable future in residential automation.

FUTURE WORK

While the current implementation of the IoT-based home automation system marks a significant achievement, there are several avenues for future exploration and enhancement:

Integration of Additional Devices: Expand the compatibility of the system to support a wider range of smart home devices, such as smart thermostats, doorbell cameras, and smart plugs, enabling users to create a more interconnected and versatile home environment.

Enhanced Automation Logic: Incorporate machine learning algorithms to develop more sophisticated automation logic based on user preferences, historical data, and environmental conditions. This could involve predictive analytics to anticipate user behavior and optimize energy usage accordingly.

Advanced Security Features: Implement advanced security features such as multi-factor authentication, biometric authentication, and anomaly detection to further strengthen the system's resilience against unauthorized access and cyber threats.

Energy Optimization Strategies: Develop energy optimization strategies using machine learning algorithms to intelligently manage energy consumption based on real-time data and user patterns. This could involve optimizing HVAC systems, scheduling device usage, and integrating renewable energy sources.

Voice Control Integration: Integrate voice control capabilities using platforms like Amazon Alexa or Google Assistant to enable users to control home devices using voice commands, enhancing accessibility and convenience.

Augmented Reality Interfaces: Explore the use of augmented reality (AR) interfaces to provide users with immersive and intuitive control over their home automation system, allowing for interactive visualization and customization of device settings.

Community Engagement and Collaboration: Foster a community of users and developers to encourage collaboration, knowledge sharing, and the development of third-party integrations and extensions for the home automation system.

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